خدمتكم عبادة نتقرب بها الى الله

2021

ملخص

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إحصاء واحتمالات

No.

* Elements of probability :-

- Definitions:

- Sample space: The set of all expected outcomes resulting from the prob. experiment denoted by Ω

e.g: In the experiment of tossing a coin one time,
find Ω

$\Omega = \{H, T\}$ → H: Head, T: tail

eg. In the experiment of tossing a coin two times,
find Ω

$\Omega = \{HH, HT, TH, TT\}$

- Event: Any subset of Ω. denoted by any letter
(A, B, C, ----)

eg let $\Omega = \{1, 2, 3\}$, Find a subset of Ω

$\{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}\{2,3\}, \{1,2,3\}, \phi$ 8 events

- Probability of event:

The probability of an event A is denoted by

$$P(A) = \frac{\#\text{ of elements in } A}{\#\text{ of elements in } \Omega}$$

Event → impossible $P(A) = 0$
→ sure $P(A) = 100\% = 1$ certain event
→ ordinary $0 < P(A) < 1$

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e.g: In the experiment of tossing a coin 2 times.
find:

a) the prob. of getting only one head.

b) the prob. of getting at least one head.

Sol:

$\Omega = \{HH, HT, TH, TT\}$

a) $A = \{HT, TH\}$ $P(A) = \frac{2}{4} = 0.5$

b) $B = \{HT, TH, HH\}$ $p(B) = 3/4 = 0.75$

e.g: In the exp of tossing a coin 3 times, Find:

a) the prob. of getting only 2 heads

b) the prob. of getting at least 2 heads.

c) the prob. of getting at most 1 head

d) the prob of getting no heads

Sol:

$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

a) $A = \{HHT, HTH, THH\}$ $P(A) = 3/8$

b) $B = \{HHT, HTH, THH, HHH\}$ $P(B) = 4/8$

c) $C = \{HTT, THT, TTH, TTT\}$ $P(C) = 4/8$

d) $D = \{TTT\}$ $p(D) = 1/8$

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e.g: In the exp. of rolling one dice, find the prob.
a) of getting an odd no.
b) of getting a prime no.

$\Omega = \{1, 2, 3, 4, 5, 6\}$

Sol: a- $A = \{1, 3, 5\}$ $P(A) = 3/6$

b- $\{1, 2, 3, 5\}$ $P(B) = 3/6$

- rolling one dice twice
=
rolling two dices once.

e.g: In the exp. of rolling two dices, find the prob. of:
a) the sum of faces is 9
b) one of the faces is twice the other
c) the 2nd entry is triple the first
d) the 1st entry is prime
e) the sum of the faces is a prime less than 4

Sol:

$\Omega = \{(1,1), (1,2), (1,3), \ldots (1,6)$
$(2,1), (2,2), (2,3) \ldots (2,6)$
$\vdots$
$(6,1), (6,2) (6,3) \ldots (6,6)\}$

$|\Omega| = 36$

a) $A = \{(3,6), (4,5), (5,4), (6,3)\}$

$P(A) = 4/36 = 1/9$

* $|\Omega|$ in the exp. of tossing a coin n times is 2^n.

* $|\Omega|$ in the exp. of rolling a dice n times is 6^n.

b) $B = \{(1,2), (2,1), (2,4), (4,2), (3,6), (6,3)\}$

$P(B) = 6/36 = 1/6$

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c) $C = \{(1,3), (2,6)\}$

$P(C) = 2/36 = 1/18$

d) $D = \{(1,1), (1,2) \dots (1,6), (2,1), (2,2) \dots (2,6), \dots (5,6)\}$

$P(D) = 18/36 = 2/3$

e) $e = \{(1,1), (1,2), (2,1)\}$

$P(D) = 3/36 = 1/12$

* Prob. rules :-

1) $P(\phi) = 0$, $P(\Omega) = 1$

2) $0 \leq P(A) \leq 1$

3) Multiplicative law of prob:

- if A, B are two independant events, then

$P(A \cap B) = P(A) \cdot P(B)$

** $A \cap B$ = A intersects B ≡ A and B ≡ both of A, B

Def : The two events are indep. if there is no relation between them and they come from the same background.

Def : The two events are called disjoint if there is no relation between them but they come from diff. backgrou

- if A, B are disjoint, then $P(A \cap B) = 0$

** Events → indep.

→ disjoint

→ dependent → $P(A \cap B) \neq 0 \neq P(A) \cdot P(B)$

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4) Additive law of prob. :

→ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

** $A \cup B \equiv A$ union $B \equiv A$ or $B \equiv$ at least one of them $\equiv$ A or B or both

** if A, B are indep.

$$P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$$

** if A, B are disjoint.

$$P(A \cup B) = P(A) + P(B)$$

e.g: Two fire detecters A, B are working independently

s.t. $P(A) = 0.2$, $P(B) = 0.4$

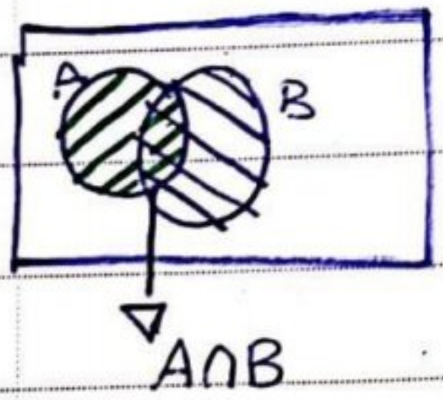
a) find the prob. that both of them will work

b) find the prob. that at least one of them will work.

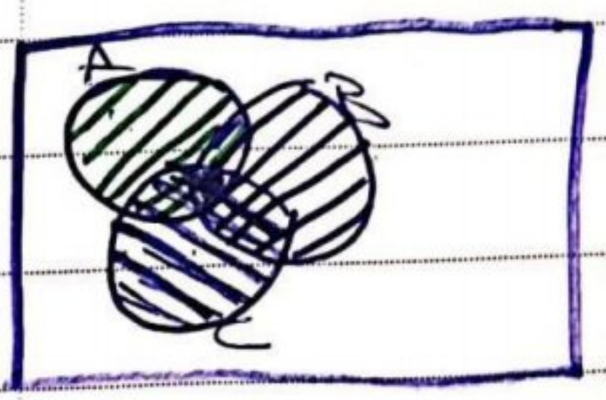
Sol:

a- $P(A \cap B) = P(A) \cdot P(B) = (0.2)(0.4) = 0.08$

b- $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.2 + 0.4 - 0.08 = 0.52$



$P(A \cup B) = P(A) + P(B) - P(A \cap B)$



$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

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e.g: A fire fighting system consists of 3 indep. component with prob. of working of:

$P(A) = 0.2$, $P(B) = 0.4$, $P(C) = 0.6$

Find the prob. that the system will work

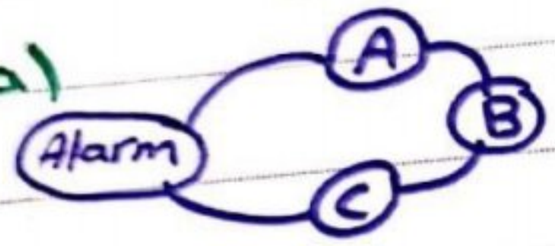
a- if the components are connected in series → indep

b- if the components are connected in parallel → dep

c- if only A is connected in series

d- if only B is connected in parallel

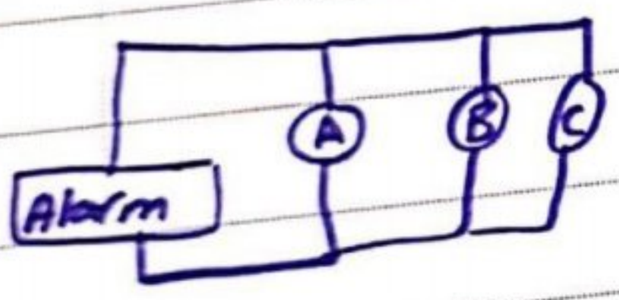
Sol: a)



$P(\text{system works}) = P(A \cap B \cap C)$

$= P(A) \cdot P(B) \cdot P(C) = (0.2)(0.4)(0.6) = 0.0$

b)



$P(\text{system works}) = P(A \cup B \cup C)$

$= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

$= 0.2 + 0.4 + 0.6 - 0.08 - 0.24 - 0.12 + 0.048 = 1.0 - 792.8$

xx $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

} → Demorgan law

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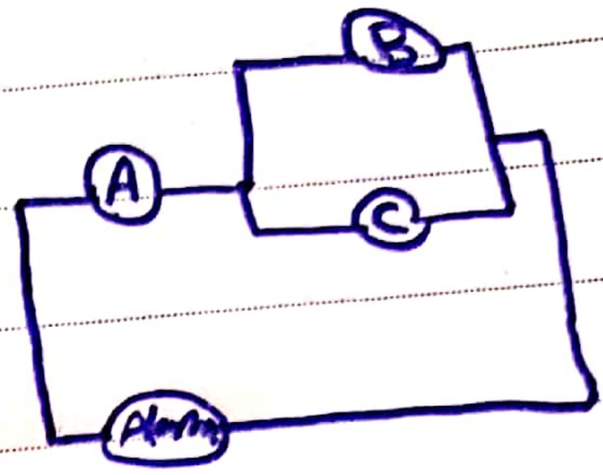
c) $P(A) = 0.2$, $P(B) = 0.4$, $P(C) = 0.6$



$P(\text{system work}) = P(A \cap (B \cup C))$

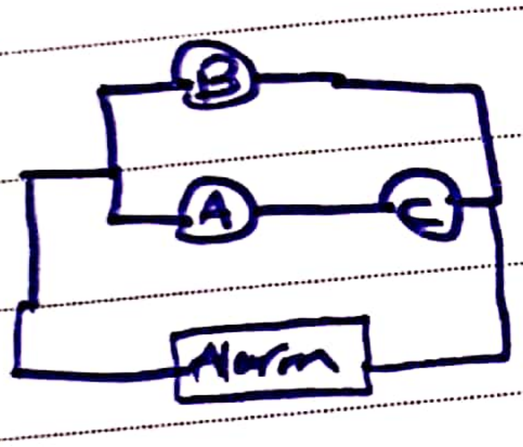
$= P(\underline{A \cap B}) \cup P(\underline{A \cap C}))$

$= P(A \cap B) + P(A \cap C) - P(\underline{(A \cap B) \cap (A \cap C)})$ → $A \cap B \cap C$

$- P(A) \cdot P(B) + P(A) \cdot P(C) - P(A) \cdot P(B) \cdot P(C)$

$= (0.2)(0.4) + (0.2)(0.6) - (0.2)(0.4)(0.6)$

d)



$P(\text{system works}) = P(B \cup (A \cap C))$

$= P(B) + P(A \cap C) - P(B \cap (A \cap C))$

$= P(B) + P(A) \cdot P(C) - P(A) \cdot P(B) \cdot P(C)$

$= 0.4 + (0.2)(0.6) - (0.2)(0.4)(0.6)$

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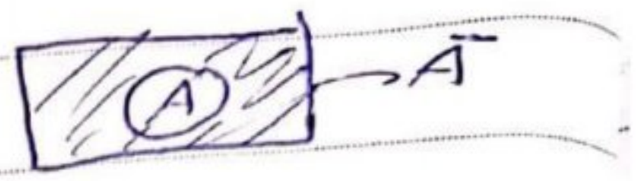
5) Complement of Prob.

$\bar{A}$ = the complement of A.

= $\Omega - A$ → All items in Ω except of A



1) $A \cup \bar{A} = \Omega$

2) $A \cap \bar{A} = \emptyset$ → disjoint

3) $P(A \cap \bar{A}) = 0$

4) $P(A \cup \bar{A}) = P(A) + P(\bar{A})$

$P(\Omega) = P(A) + P(\bar{A})$

$1 = P(A) + P(\bar{A})$

$P(\bar{A}) = 1 - P(A)$

Theorem: If A, B are indep., then

1) $A, \bar{B}$ are indep.

2) $\bar{A}, B$ " "

3) $\bar{A}, \bar{B}$ " "

Proof:

3) we want to show that $\bar{A}, \bar{B}$ are indep.

i.e. we want to show that

$P(\bar{A} \cap \bar{B}) = P(\bar{A}) \cdot P(\bar{B})$

L.H.S

$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B})$

$= 1 - P(A \cup B)$

$= 1 - [P(A) + P(B) - P(A \cap B)]$

$= 1 - P(A) - P(B) + P(A) \cdot P(B)$

$= 1 - P(B)(1 + P(A))$

الدكتور ما بده الاثبات

"عشان التعلم عن بعد"

smile

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$= P(\bar{A}) - P(B)[1 - P(A)]$ (2)

$P(\bar{A}) - P(B)\,P(\bar{A})$

$= P(\bar{A})[1 - P(B)]$

$= P(\bar{A}) \cdot P(\bar{B})$ #

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* **Demorgan's laws** :

1) $\bar{A} \cap \bar{B} = \overline{A \cup B}$
2) $\bar{A} \cup \bar{B} = \overline{A \cap B}$
3) $\overline{A \cap B \cap C} = \bar{A} \cup \bar{B} \cup \bar{C}$
4) $\overline{A \cup B \cup C} = \bar{A} \cap \bar{B} \cap \bar{C}$

Ex A, B are indep. Components such that

$P(A \text{ works}) = 0.3$, $P(B \text{ works}) = 0.6$

Find the prob. that:

a) B will fail.

b) at least one of them will fail.

c) A will fail & B will fail

d) both of them will fail

Sol:

a) $P(B \text{ fail}) = P(\bar{B}) = 0.4$

b) $P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B})$

$= 1 - P(A \cap B)$

$= 1 - P(A) \cdot P(B)$

$= 1 - (0.3)(0.6) = .82$

$\Rightarrow$ ...

smile for life

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No.

$\overline{A \cup B}$

c* ~~$A \oplus B$~~ will fail

~~$P(\overline{A \cup B})$~~ $= P(\bar{A} \cap \bar{B})$

$= P(\bar{A}) \cdot P(\bar{B})$ $= (0.7)(0.4)$

~~d) $P(\overline{A \cap B})$~~ $= 1 - P(A \cap B)$

$= 1 - P(A) \cdot P(B)$

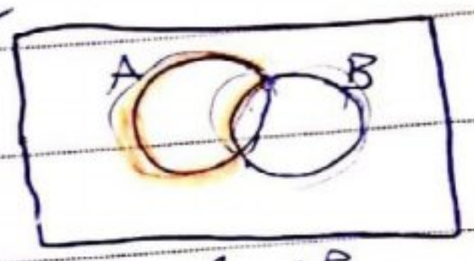
$= 1 - (0.3)(0.6)$

6 Exception law Prob. :- قانون الطرح

$P(A - B) = P(A \cap \bar{B})$

$= P(A) - P(A \cap B)$ بنوخذ P(A) عامل مشترك

$\sim P(\bar{A} \cap B) = P(B) - P(A \cap B)$



مثل سي if
عنطقت التقاطع

* IF A,B are indep. , then

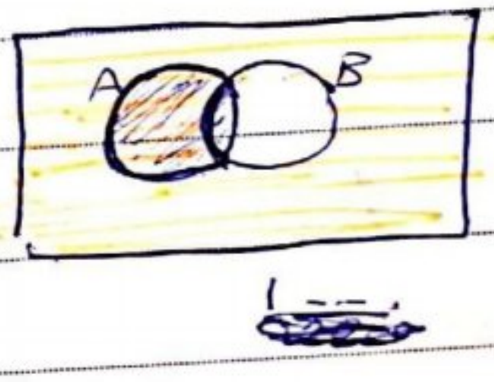
$P(A \cap \bar{B}) = P(A) - P(A \cap B)$

$= P(A) - P(A)\,P(B)$

$= P(A) \cdot (1 - P(B))$

$= P(A) \cdot P(\bar{B})$

الخطوة الي
تحتوي على نوعين
من الخطوط



لو كان عنا مجموعة تقاطع مجموعة ثانية Complement بتكون هي بعد المجموعة لي ما عليها Complement ناقص الي تحت Complement

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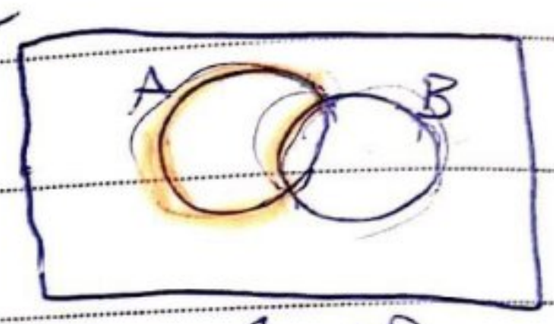
6 Exception law prob. ٥- قانون الطرح

$$P(A-B) = P(A \cap \bar{B})$$
$$= P(A) - P(A \cap B)$$
$$\sim P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

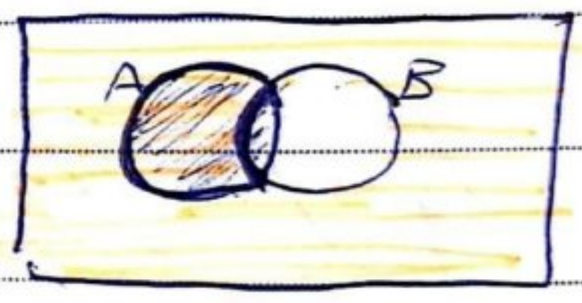


المنطقة المظللة هي

- IF A,B are indep., then

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$
$$= P(A) - P(A)\,P(B)$$
$$= P(A).(1 - P(B))$$
$$= P(A) \cdot P(\bar{B})$$

المنطقة التي تحتوي على توعين من الاحتمال



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Ex1 If $P(A) = 0.2$, $P(\bar{A} \cap B) = 0.3$,
find $P(A \cup B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= P(A) + P(\bar{A} \cap B)$$
$$= 0.2 + 0.3 = 0.5$$

Ex2 if $P(A) = 0.4$, $P(A \cap \bar{B}) = 0.1$
Find $P(\bar{A} \cup \bar{B})$

$$P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B})$$
$$= 1 - P(A \cap B)$$

But $P(A \cap \bar{B}) = P(A) - P(A \cap B)$
$0.1 = 0.4 - P(A \cap B)$
$P(A \cap B) = 0.3$

So, $P(\bar{A} \cup \bar{B}) = 1 - P(A \cap B)$
$= 1 - 0.3 = 0.7$

Ex3 If $P(A \cup B) = 0.9$, $P(A) = 0.7$, $P(B) = 0.4$ Find $P(A \cup \bar{B})$

$0.9 = P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$0.9 = 0.7 + 0.4 - P(A \cap B)$
$\rightarrow$ $P(A \cap B) = 0.2$

Now; $P(\overline{A \cup B}) = P(\bar{A} \cap \bar{B}) = P(\bar{A} \cap \bar{B})$
$= P(B) - P(A \cap B)$
$= 0.4 - 0.2 = 0.2$

مشطوب

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قانون الشرط : احتمالية اشي شرط اشي ثاني يساوي احتمالية التقاطع على احتمالية الشرط

7 Conditional probability: الاحتمال بشرط

~ P(A), P(B), P(C) ... Marginal prob. الاحتمال اللذي يعبر احتمال حادث واحد (هامشي).

~ P(A∩B) ≡ P(AB) = P(ABC), P(AC), ... → Joint Prob. احتمال حدوث حادثين.

~ P(A|B), P(B|C), P(A|C), ... → Conditional Prob.
given

* P(A|B) = Pro. of A given B = Prob. of event A s.t event B is done

نقرأ احتمال حادث الذي يقع خلق P بشرط حدوث الثاني.

↝ If A, B are indep. then لا يؤثر بعض

$$P(A|B) = P(A)$$

هاي الشحطة تعبر عن الـ indp

* $P(A|B) = \frac{P(A \cap B)}{P(B)}$ في المقام يكون احتمال وشرط ↳ $P(A \cap B) = P(A|B) \cdot P(B)$

$P(B|A) = \frac{P(B \cap A)}{P(A)}$ → $P(A \cap B) = P(B|A) \cdot P(A)$

↝ If A, B are indep. then:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B)}{P(B)} = P(A)$$

الشرط بنشطب لنه ما اله معنى بسبب انه ما اله علاقة على الـ A

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نستطيع معرفة لو كانت الاحتمالية لا تعتمد على الشرط عن طريق جواب ضرب الاحتمالية الأولى بالشرط مثلا لو ضربنا بالمثال الاتي
P(A) x P(B) الجواب 0.2" وهيك بعرف غير معتمدين عن بعض عنه لجواب بساوي ناتجهم

Ex] In some population, it's found that
$P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.2$
where A denotes to the hypertension ارتفاع ضغط الدم
& B denotes to the over weight
If a person is chosen at random, find the prob. that he has a hypertension Sol! he has an over weight.

Sol:
$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.5} = 0.4$$

حادثين مستقلين

$P(A|B) = P(A)$, So, A is indep of B كحادثين منفصلين

If $P(A) = 0.2$ → لو كانت ~ → يكونو بينهم علاقة → dep.

* Rules:

1) $P(\phi|A) = 0$
2) $P(\Omega|A) = 1$
3) $P(\bar{A}|B) = 1 - P(A|B)$

احتمالية خاي بدون شرط

Proof:

$$P(\bar{A}|B) = \frac{P(\bar{A} \cap B)}{P(B)}$$

$$= \frac{P(B) - P(A \cap B)}{P(B)}$$ ← وزعنا البسط على المقام

$$= 1 - \frac{P(A \cap B)}{P(B)}$$ الاحتمال الشرطي

$$= 1 - P(A|B)$$ #

4) $P(A|\bar{B}) \neq 1 - P(A|B)$

$$P(A|\bar{B}) = \frac{P(A \cap \bar{B})}{P(\bar{B})}$$

$$= \frac{P(A) - P(A \cap B)}{1 - P(B)}$$

مشطوب

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No.

* permutations and combinations:- التباديل والتوافيق والتجاميع

* No. of ways of chosing r elements out of n elements s.t. the order is important is denoted by

$$nP_r = \frac{n!}{(n-r)!}$$

* No. of ways of chosing r elements out of n s.t the order is not important is denoted by

$$nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Ex| How many 3 digits car plates could you form from $\{0,1,2,\dots,9\}$

→ No. of plates $= (10)^3$

Ex| How many car plates could you form from $\{0,1,\dots,9\}$ without duplicated any number

بدون تكرار

no. of plates $= 10 \times 9 \times 8 = \boxed{10P3} = \frac{10!}{(10-3)!} = \frac{10!}{7!} = \frac{10\times9\times8\times7!}{7!}$

= Chosing 3 out of 10 without replacement

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التربيعي هو نفسه سحب كرتين بدون ارجاع.

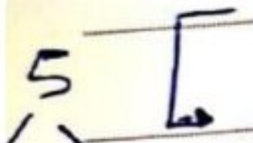
No.

Ex) In the exp. of drawing two balls from a box with 3 red and 2 black balls,

a) Find no. of ways of choosing 1 red ball التربيعي غير مهم
b) find no. of ways of Choosing 2 red balls.
c) find no. of ways of Choosing 2 balls.
d) find the prob. of Choosing 1 red ball
e) find the prob. of Choosing 2 red balls.

Sol :-

$\Omega = \{R_1R_2, R_1R_3, R_2R_3, R_1B_1, R_1B_2, R_2B_1, R_2B_2, R_3B_1, R_3B_2, B_1B_2\}$



a) No. of way of choosing 1 red ball

is $= \binom{3}{1}\binom{2}{1} = 6$

3 choose 1

b) No. of ways of choosing 2 red

$= \binom{3}{2}\binom{2}{0} = 3$

c) No. of ways of choosing 2 balls $= \binom{5}{2} = 10 = \frac{\#}{den}$

d) $p(1\ red) = \frac{\#(1\ red)}{\#(\Omega)} = \frac{\binom{3}{1}\binom{2}{1}}{\binom{5}{2}} = 0.3$

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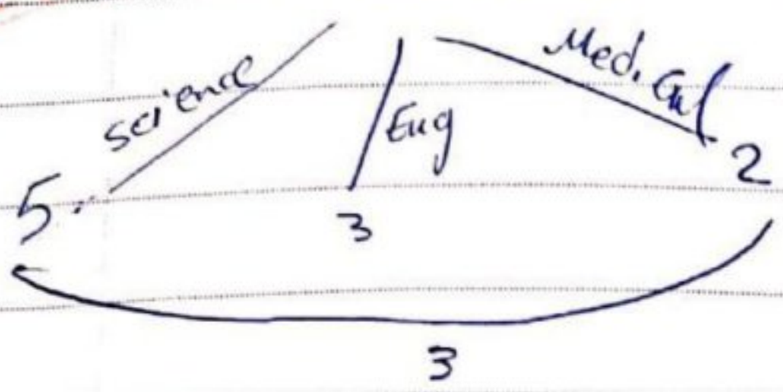
No.

e) $P(2\text{ red}) = \frac{\#(2\text{ red})}{\#\Omega} = \frac{\binom{3}{2}\binom{2}{0}}{\binom{5}{2}} = 0.3$

[٧] 10 students — بدّه اختار 3 طلاب عشوائيًا



a) Find the prob. of choosing 2 sci student.

$$P(2S) = \frac{\binom{5}{2}\binom{5}{1}}{\binom{10}{3}}$$

b) Find the prob. of choosing at least 2 Eng. students

٣ أو ٢ إنجليزي

$$P(\text{at least } 2E) = P(2E) + P(3E)$$

$$= \frac{\binom{3}{2}\binom{7}{1}}{\binom{10}{3}} + \frac{\binom{3}{3}\binom{7}{0}}{\binom{10}{3}}$$

c) Find the prob. of choosing 1 sci. and at least 1 Eng.

$$P(1S \cap \text{at least } 1E) = P(1S \cap \text{at } 1E) + P(1S \cap 2E)$$

$$= \frac{\binom{5}{1}\binom{3}{1}\binom{2}{1}}{\binom{10}{3}} + \frac{\binom{5}{1}\binom{3}{2}\binom{2}{0}}{\binom{10}{3}}$$

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No.

d) Find the prob. of chosing at least 1 Medical Student

$$P(\text{at least } 1M) = P(1M) + P(2M)$$

$$= \frac{\binom{2}{1}\binom{8}{2}}{\binom{10}{3}} + \frac{\binom{2}{2}\binom{8}{1}}{\binom{10}{3}}$$

* Random Variables:-

Def: The Random Variables (r.v) is a function defined from Ω to $\mathbb{R}$.

Range

i.e $X: \Omega \longrightarrow \mathbb{R}$ → real no. ج

الأعداد

~ The Range of the r.v X is denoted by R_X which is the set of all possible values of the r.v X.

e.g) X : No. of student who pass the exam o

$X = \{0, 1, \dots, 10\} = R_X$

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No.

* The r.v's have two main types?

(I) Discret r.v : الفيزياء

↳ It takes values as distinct no.'s.

e.g. Y: No. of heads in the exp. of tossing a coin 5 times

$Y = \{0, 1, 2, \ldots, 5\}$

(II) Cont. r.v : الفيزياء

It takes values as intervals.

e.g. Z: The weight of boxing championship.

$Z = \{(49, 52), (52, 56), \ldots, (91, 206)\}$.

Z_1 Z_2

Def:

The prob. density function (p.d.f): is denoted by f (maybe f(x) in case of cont. r.v's) which is prob. function satisfies the following :-

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Cont. / discrete كالسّ في pdf لـ حواجب *

↳ discrete:

1) $P(x) \geq 0$, $\forall x \in R_X$

2) $\sum_{x \in R_X} P(x) = 1$ مجموعة من النقاط تجمع

3) $P(X \in A) = \sum_{x \in A} P(x)$, where $A \subseteq R_X$

~~4) $P(X \in A) \cap (X \in B) = P(X \in A \cap B) = \sum_{x \in A \cap B} P(x)$~~

5) $P(X \in A | X \in B) = \frac{P(X \in A \cap B)}{P(X \in B)}$

↳ Cont. :-

1) $f(x) \geq 0$, $\forall x \in R_x$

لـ interval فترة

2) $\int_{R_X} f(x)\,dx = 1$ تتكون من فترة او عدد لانهائي من الفترات

3) $P(X \in A) = \int_{(A)} f(x)\,dx$

فترة او اتحاد فترات

4) $P(a \leq x \leq b) = \int_a^b f(x)\,dx$

5) $P(X = a) = \int_a^a f(x)\,dx = 0$

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↳ discrete :-

6) The expected value of the r.v X is denoted by:

$$EX = \sum_{x \in R_x} x\,p(x)$$

7) $EX^2 = \sum_{x \in R_x} x^2 p(x)$

$EX^3 = \sum_{x \in R_x} x^3 p(x)$ / $Eg(x) = \sum_{x \in R_x} g(x)\,p(x)$

8) $E(aX+b) = aEx + b$ / $E(ax^2+bx+c) = aEx^2 + bEX + C$

9) $\text{Var } X = \sigma_x^2 = EX^2 - (EX)^2$

10) $\text{Var}(ax+b) = a^2 \text{Var} X$

↳ Cont. :

مطلوب

6) $P(x \in A \cap x \in B) = \int_{A \cap B} f(x)\,dx$

7) $P(x \in A \mid x \in B) = \dfrac{P(x \in A \cap B)}{P(x \in B)} = \dfrac{\int_{A \cap B} f(x)\,dx}{\int_B f(x)\,dx}$

8) $EX = \int_{R_x} x\,f(x)\,dx$

9) $EX^2 = \int_{R_x} x^2 f(x)\,dx$ / $EX^3 = \int_{R_x} x^3 f(x)\,dx$ / $Eg(x) = \int_{R_x} g(x) f(x)\,dx$

10) $E(aX+b) = aEx + b$ / $E(ax^2+bx+c) = aEx^2 + bEx + C$

11) The variance of the r.v. X is

$\text{Var } X = \sigma_x^2 = EX^2 - (EX)^2$

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No.

الانحراف المعياري
Standard deviation

Discrete:

11) $std(x) = \sqrt{\sigma_x^2}$ → varian

12) $std(ax+b) = |a|\, std(x)$ → الـ B يروح

مثال:
لو قدمنا امتحان وكان في خطا واجا الدكتور رفعنا كلنا
10 علامات شو بأثر بتباين العلامات ؟
ولا شي بنهضلزي هو

Standard deviation
لو ربعناه بعطيني الـ **variance**

Cont:

12) $Var(aX+b) = a^2 VarX$

13) $std(x)$ = standard deviation $= \sqrt{\sigma_x^2}$

14) $std(ax+b) = |a|\, std(x)$.

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#Fida'a_Abbadi

No.

11/10/2017

* EX ≡ expectation of X
≡ mean of x = average of x ≡ $\bar{x}$ المتوسط الحسابي

~ If $y = ax + b$, then

1) $Ey = E(ax+b) = aEX + b$

$\bar{y} = a\bar{x} + b$

2) $Var\ y = Var(ax+b)$

$= a^2\ Var\ X$.

لا يتأثر الثابت بعامل الضرب

يتأثر x بمتغير

e.g:-

R.V. X

$y = x + 5$

$\sigma^2_x = \sigma_y$

الانحراف لا يتأثر بالقيمة الثابتة

$z = 5x$

$\sigma^2_z > \sigma^2_x$

إذا ضربت r.v x

برقم أكبر من 1 يزداد

الانحراف يتأثر بالقيمة

واحد أقل من واحد تقل الانحرافات.

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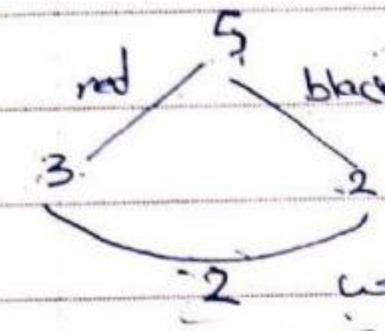
* $\binom{n}{r} = \binom{n}{n-r}$ $\binom{10}{1} = \binom{10}{9}$ $\binom{10}{2} = \binom{10}{8}$ ...

No.

Ex) Two balls are drawn from a box with 3 red and 2 black balls. let x be the no. of red balls. Find every things:



$X = \{0, 1, 2\}$

احتمال اصحب ←

$P(X=0) = P(0) = \frac{\binom{3}{0}\binom{2}{2}}{\binom{5}{2}} = 0.1$

$P(X=1) = P(1) = \frac{\binom{3}{1}\binom{2}{1}}{\binom{5}{2}} = 0.6$

$P(X=2) = P(2) = \frac{\binom{3}{2}\binom{2}{0}}{\binom{5}{2}} = 0.3$

△ * $\sum_{x \in R_x} P(x) = P(0) + P(1) + P(2) = 1$

X	0	1	2	
P(x)	0.1	0.6	0.3	1

← Prob. distribution table

جدول توزیع احتمالی ← مجموع احتمال

discrete

X	P(x)	x P(x)	$x^2 P(x)$	$x^3 P(x)$
0	0.1	0	0	0
1	0.6	0.6	0.6	0.6
2	0.3	0.6	1.2	2.4
مجموع	1	1.2	1.8	3

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No.

$\rightsquigarrow Ex = \sum_{x \in Rx} x p(x) = 1.2$

$\rightsquigarrow Ex^2 = \sum_{x \in Rx} x^2 p(x) = 1.8$

$Ex^3 = \sum_{x \in Rx} x^3 p(x) = 3$

$\rightsquigarrow$ $\text{Var } X = Ex^2 - (Ex)^2 = 1.8 - (1.2)^2 = 0.36$

$\rightsquigarrow$ $\text{std}(x) = 0.6 = \sqrt{0.36}$

- If $y = -2x + 3$, then

$Ey = E(-2x+3) = -2Ex + 3$

$= (-2)(1.2) + 3 = 0.6$

$\rightsquigarrow$ $\text{Var } y = \text{Var}(-2x+3) = 4 \text{ var } x = (4)(0.36) = 1.44$

$\rightsquigarrow$ $\text{std}(y) = \text{std}(-2x+3) = |-2| \text{ std}(x)$

$= (2)(0.6) = 1.2$

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You can do it

No.

ما الـ علاقة اذا كانة Can او dis.

Ex) let X be a r.v. with $EX = 2$, $\text{Var}X = 3$
find $E(-2X^2 + 3X - 5)$

Sol:

$$3 = \text{Var}X = EX^2 - (EX)^2$$
$$3 = EX^2 - (2)^2 \longrightarrow EX^2 = 7$$
$$\text{So, } E(-2X^2 + 3X - 5)$$
$$= -2EX^2 + 3EX - 5$$
$$= (-2)(7) + (3)(2) - 5 = -13$$

Ex): let X be a discrete r.v with range $R_X = \{1, 2, 3\}$ s.t $EX = 2$, $\text{Var}X = 1.3$, find $p(X \in A)$; $A = \{1, 3\}$.

Sol: $EX^2 = \text{Var}X + (EX)^2$

$EX^2 = 1 + 4 = 5$

$2.4 = EX = \sum_{x \in R_X} x\,p(x)$

$5.72 = EX^2 = \sum_{x \in R_X} x^2 p(x)$

$5.72 = (1)p(1) + 4p(2) + 9p(3)$ — (2)

$\sum_{x \in R_X} p(x) = 1$

$p(1) + p(2) + p(3) = 1$ — (3)

محلول غلط الدكتور حله بالصفحات الجاي

الفكرة: انه احنا بنعوف توقع X^2 فنستخدم الحذف والتعويض عشان نطلع احتمالية {1,2,3}

(1)

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You strong enough

No.

$P(1) = 0.5$, $P(2) = 0$, $P(3) = 0.5$

$P(x \in \{1,3\}) = P(1) + P(3) = 0.5 + 0.5 = 1$

Ex) let X be a discrete r.v with p.d.f

$$P(x) = \begin{cases} c, & x = 1, 2, 3, 4 \\ 0.1, & x = 5, 6 \\ 0, & o.w \end{cases}$$

Find c.

Sol:

$$\sum_{x \in R_X} p(x) = 1$$

$$P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 1$$

$$C + C + C + C + 0.1 + 0.1 = 1$$

$$4C = 0.8 \qquad \boxed{C = 0.2}$$

Ex): let x be a Cont. r.v with p.d.f

$$F(x) = \begin{cases} c, & 0 \le x < 1/2 \\ 2c, & 1/2 < x \le 1 \\ 0, & o.w \end{cases}$$

Find c.

Sol:

$$\int_{R_X} f(x)\,dx = 1 \longrightarrow \int_0^{1/2} c\,dx + \int_{1/2}^{1} 2c\,dx = 1$$

$$\tfrac{1}{2}c + 2\left(\tfrac{1}{2}c\right) = 1$$

$$\tfrac{1}{2}c + c = 1$$

$$\tfrac{3}{2}c = 1 \qquad \boxed{c = \tfrac{2}{3}}$$

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Eg. X discrete r.v $x \in [1,2,3]$

$E(X) = 2$, $Var(X) = 1$

find $P(x \in A)$ where $A = [1,3]$

Sol.

$$Var(X) = E(X^2) - (E(X))^2$$

$$\rightarrow 1 = E(X^2) - 2^2$$

$$\rightarrow E(X^2) = 5 .$$

Since $2 = E(X) = \sum_{x \in [1,2,3]} x\,p(x) = (1)\,p(1) + (2)\,p(2) + 3\,p(3) = 2$

$$\rightarrow \boxed{p(1) + 2p(2) + 3p(3) = 2} \quad (1)$$

Since $E(x^2) = 5$, hence

$$E(x^2) = \sum x^2 p(x) = \boxed{p(1) + 4p(2) + 9p(3) = 5} \quad (2)$$

Since $\sum p(x) = 1 \rightarrow \boxed{p(1) + p(2) + p(3) = 1} \quad (3)$

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by solving ①, ② & ③, we have

$p(1) = \frac{1}{2}$, $p(2) = 0$, $p(3) = \frac{1}{2}$

$\rightarrow p(x \in \{1,3\}) = p(1) + p(3)$

$= \frac{1}{2} + \frac{1}{2} = 1$

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No.

Ex) let X be a Cont. r.v with p.d.f :-

$$f(x) = \frac{3}{2}C^2 e^{-cx}, \quad 0 < x < \infty$$

a) Find C

$$\int_{Rx} f(x)\,dx = 1 \longrightarrow \int_0^\infty \frac{3}{2}C^2 e^{-cx}\,dx = 1$$

$$\longrightarrow c^2 \frac{-1}{c} e^{-cx}\Big]_0^\infty = 1$$

C عدد موجب

$$-c\, e^{-cx}\Big]_0^\infty = 1 \qquad -c[0-1] = 1 \longrightarrow \boxed{C = +1}$$

b) find $P(X \in (0,1))$

$$P(X \in (0,1)) = P(0 < x < 1)$$

$$= \int_0^1 f(x)\,dx = \int_0^1 e^{-x}\,dx = -e^{-x}\Big]_0^1$$

$$= -(e^{-1} - e^{0}) = 1 - 1/e$$

$f(x) = e^{-x}$

احتمالية اكس تنتمي لفترة هي التكامل على الفترة

- $P(X \in (-1,2) \cap X \in (1,15))$

$$= P(X \in (1,2)) = \int_1^2 e^{-x}\,dx = \text{as part b.}$$

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No.

16/10/2017

c) Find $P(X=2)$

$$P(X=2) = \int_2^2 f(x)\,dx = 0.$$

كس نس اوي عدد احتماليتهابتصيد 0 لنلاون ص24 اخر سطر

d) Find EX

$$Ex = \int_{R_x} x\, f(x)\,dx = \int_0^\infty x\, e^{-x}\,dx$$

$$= -x e^{-x} - e^{-x} \Big]_0^\infty = 0 - (0-1) = 1$$

e) Find Ex^2

$$Ex^2 = \int_0^\infty x^2 e^{-x}\,dx \quad (*) \qquad (*)$$

$$= -x^2 e^{-x} - 2x e^{-x} - 2e^{-x} \Big]_0^\infty$$

$$= 0 - (0 - 0 - 2) = 2$$

f) Find $E(2x^2 - 3x + 5)$

$$E(2x^2 - 3x + 5)$$

$$= 2Ex^2 - 3Ex + 5$$

$$= (2)(2) - (3)(1) + 5 = 6$$

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g) Find Var(x)

$$\text{Var}X = EX^2 - (EX)^2 = 2 - (1)^2 = 1$$

h) Find $E(2X-3)$, $\text{Var}(2X-3)$

$$E(2X-3) = 2EX - 3$$
$$= (2)(1) - 3 = -1$$

$$\text{Var}(2X-3) = (2)^2 \text{Var}X = (4)(1) = 4.$$

Ex) let x be a Cont. r.v with p.d.f

$$f(x) = \begin{cases} C, & 0 \le x < 1 \\ 2C, & 1 \le x < 2 \\ 0, & \text{o.w} \end{cases}$$

H.W

a) Find c b) Find $P(X=1)$

c) Find $P(\frac{1}{2} < x < 1.5)$

d) Find EX, EX^2, $\text{Var}X$

e) Find $E(2-3x)$, $\text{Var}(2-3x)$.

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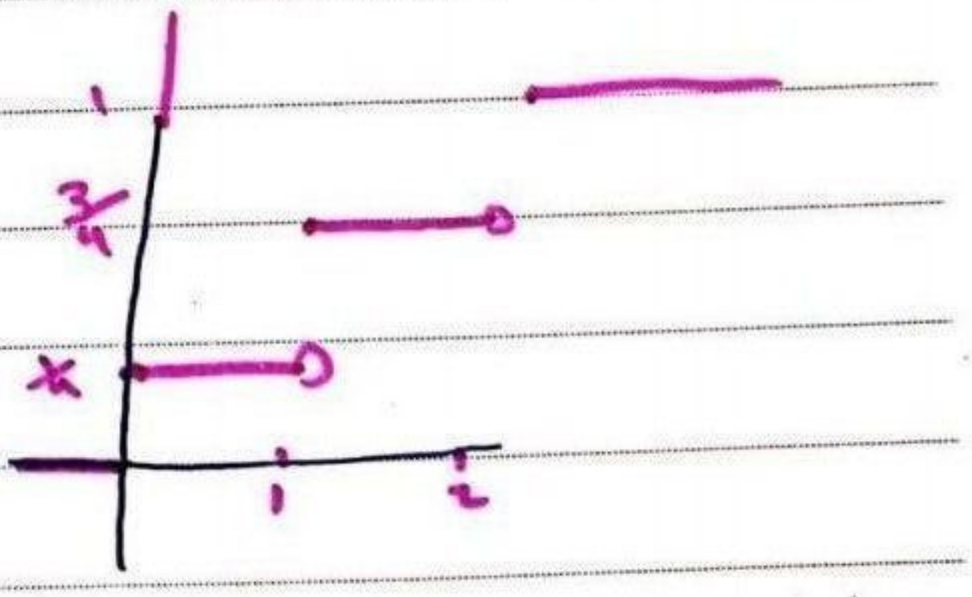
Def: let X be a r.v. with range R_x, then the Cumulative distribution function (c.d.f) ← اقتران التوزيع التراكمي or Simply the distribution function (d.f) is given by

$$F(x) = \begin{cases} \sum_{u \le x} p(u) & ; \ X \text{ is discrete} \\ \int_{-\infty}^{x} f(u)\,du & ; \ X \text{ is Cont.} \end{cases}$$

هون حطينا حرف U لنه ما بصير يكون حد التكامل جوا القوسين f(x)

Ex) let X be a discrete r.v with the following:-

X	0	1	2
P(x)	1/4	1/2	1/4



a) Find the c.d.f
b) Find $P(1 \le x < 2)$
c) Find $p(0.5 < x < 1.5)$

Sol:

a) $F(x) = 0 \ ; \ x < 0$

$F(x) = 1/4 \ ; \ 0 \le x < 1$

$F(x) = \frac{1}{4} + 1/2 \ ; \ 1 \le x < 2$

$F(x) = \frac{1}{4} + 1/2 + \frac{1}{4} \ ; \ 2 \le x$

$$\rightarrow F = \begin{cases} 0 & ; \ x < 0 \\ \frac{1}{4} & ; \ 0 \le x < 1 \\ \frac{3}{4} & ; \ 1 \le x < 2 \\ 1 & ; \ x \ge 2 \end{cases}$$

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No.

b) $p(1 \leq x \leq 2) =$

$= p(1) + p(2) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$

علامة اليساوي مهمة بالدسكريت
بس مش مهمة بالـ cont

c) $p(0.5 < x < 1.5) =$

$= p(1) = \frac{1}{2}$

* Properties:-

1) $F(x)$ is a step function when x is discrete

2) $F(x)$ is Cont. when x is Cont.

3) $F(-\infty) = 0$, $F(\infty) = 1$

4) $p(a \leq x \leq b) = F(b) - F(a)$ if x Cont. r.v

5) $f(x) = \frac{d}{dx}(F(x))$; if x is Cont. cdf is pdf. cont.

6) $F(x_1) \leq F(x_2)$; if $x_1 \leq x_2$

i.e. F is increasing.

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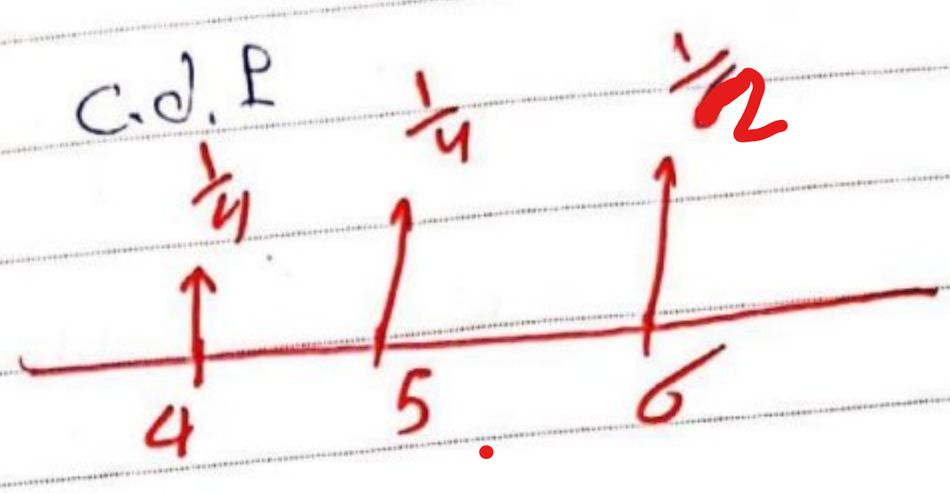
No.

Ex) let x be a discrete r.v with c.d.f

$$F(x) = \begin{cases} 0 & x<4 \\ 1/4 & 4 \le x<5 \\ 1/2 & 5 \le x<6 \\ 1 & x \ge 6 \end{cases}$$



لو فاهمين طريقة الحل
مافي داعي للتفصيل

a) Find a p.d.F
b) Find $P(4 \le x \le 6)$

طريقة إيجاد p.d.F

a) $P(X=4) = F(4) = 1/4$

$P(x=5) = F(5) - F(4) = 1/2 - 1/4 = 1/4$

$P(X=6) = F(6) - F(5) = 1 - 1/2 = 1/2$

$F(4.5) = 1/4$ / $P(X=4.5) = 0$ ← لأنه لازم عند نقطة معينة وحتى ارتفاع الدرجة عند الدرجة اللي قبلها

ال p.d.F عند حدوث ← عند ارتفاع الدرجة
وال c.d.F ← عند الدرجة نفسها

b) $P(4 \le x \le 6) = P(4) + P(5) + P(6)$
$= 1/4 + 1/4 + 1/2$
$= 1$

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Ex) let x be a Cont. r.v. with p.d.f

$$f(x) = e^{-\lambda x}, \quad x \in (0, \infty)$$

Find. a) The c.d.f

b) $P(1 \leq x \leq 2)$

c) $P(1 < x < 2)$

Sol:

a) $$F(x) = \int_{-\infty}^{x} f(u)\,du = \int_{0}^{x} e^{-\lambda u}\,du$$

$$= \frac{-1}{\lambda} e^{-\lambda u} \Big]_0^x = \boxed{\frac{-1}{\lambda}\left[e^{-\lambda x} - 1\right]}$$

b) $$P(1 \leq x \leq 2) = P(1 < x < 2)$$

$$= \int_1^2 f(x)\,dx$$

or

$$= F(2) - F(1)$$

$$= -\frac{1}{\lambda}\left[e^{-2\lambda} - 1\right] + \frac{1}{\lambda}\left[e^{-\lambda} - 1\right]$$

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ملاحظة: لما يكون الاف كبيتل F بكون CDF و r.v cont ——— No. ——— CDF (F)

طرقة الحل عشان اوجد الpdf بشتق الcdf

Ex) let $F(x) = \ln(1-\alpha x)$,

find $f(x)$.

a) $f(x) = \frac{d}{dx}(F(x))$

$= -\frac{-\alpha}{1-\alpha x}$

$= \frac{\alpha}{1-\alpha x}$

b) Find $P(0<x<0.5) =$

$P(0<x<0.5) = \int_0^{0.5} f(x)\,dx$

$= \int_0^{0.5} \frac{\alpha}{1-\alpha x}\,dx$

or $= F(0.5) - F(0)$

$= 1 - \ln(1-0.5\alpha) - (1 - \ln(1))$

$= -\ln(1-0.5\alpha)$.

c) Find $P(X=0.25) = 0$

smile for life

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Wensday

18-10-2017

No.

* Bivariate random Variables :-

عنا 2 ر**r.v** ممكن يكونو ال2 **Discrete** او 2 **cont** وممكن يكون حالة تالتة بس مش مطلوب منا غير الحالتين

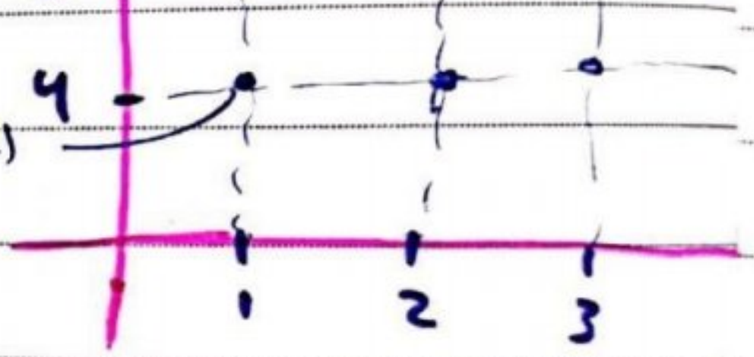
Def:- let x, y be two discrete r.v's with ranges R_x, R_y, then the two dimensional range of x, y is given by

هون راح نتعامل مع ازواج مرتبة

$$R_{xy} = \{(x,y) \; ; \; x \in R_x, \; y \in R_y\}$$

الرسمات مش كثير مهمة

discrete



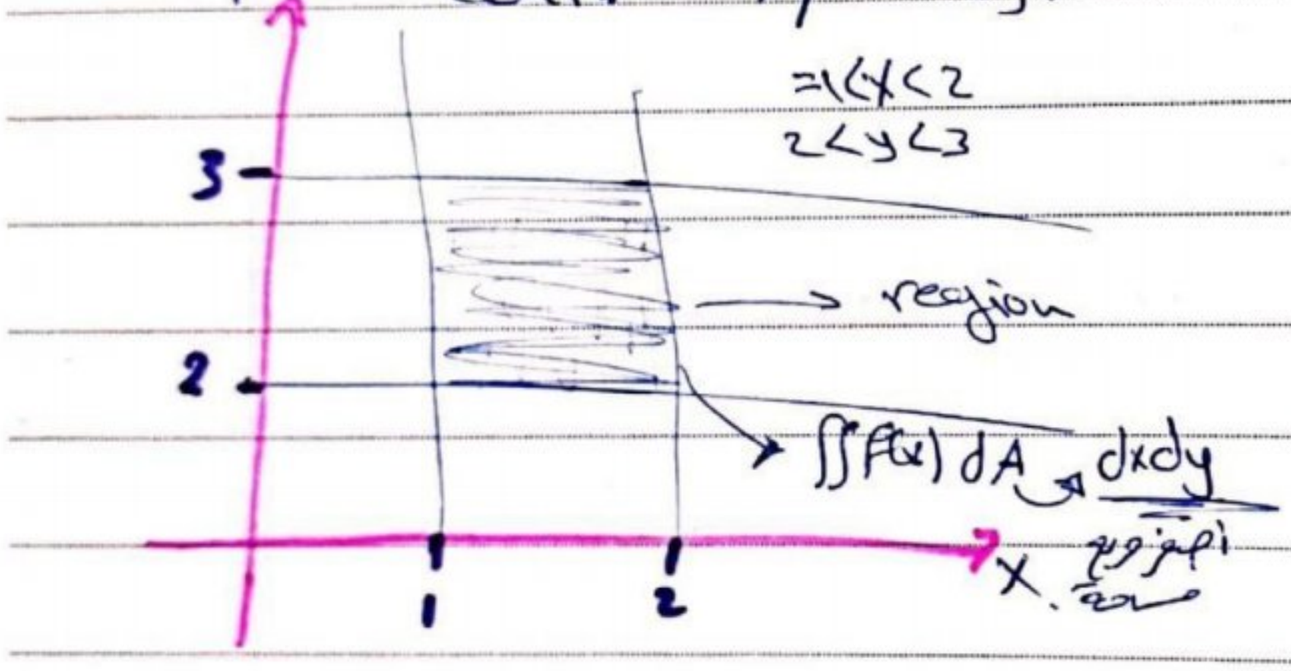
نقاط x/y الـRang

هي وحدة نتعامل معها هي النقطة

cont. $R_{xy} = \{(x,y)$

$1<x<2$

$2<y<3$



أصغر وحدة مساحة نتعامل معها هي وحدة واحدة

$(dx) = \lim \Delta x$

يتكامل على أصغر وحدة - حولها يقترب عند الصفر.

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بستخدم "joint" عشان
اجمع بين p.d.f تبع
X & Y

- with joint p.d.f :-

$$f(x,y) = P_{\text{R}}\{X=x \cap Y=y\}$$

which satisfies the following :-

بنعوض كل النقاط بالاقتران X & Y
وبنجمع الأجوبة هاي بحالة Discrete
اما بالـ cont بصير دبل انتقريشن بدنا
نكامل بالنسبة لـ X وبالنسبة لـ Y

1) $f(x,y) \geq 0, \forall (x,y) \in R_{xy}$.

لـ في حالة
تقطع
Region

2) $\sum_{x\in R_x}\sum_{y\in R_y} f(x,y) = 1$, $\iint_{R_{xy}} f(x,y)\,dx\,dy = 1$

$(x,y)\in R_{xy}$

cont.

3) The marginal p.d.f. of X is given by

$$f_x(x) = \sum_{y\in R_y} f(x,y).$$

cont.

$$f_x(x) = \int_{R_y} f(x,y)\,dy$$

$$f(x,y) = x+y$$
$$\int_0^1 (x+y)\,dy$$
$$= xy + \frac{y^2}{2}\Big|_0^1$$
$$= x + \frac{1}{2} - 0$$

هون بس نكامل بالنسبة لـ X
بنعتبر الـ Y ثابت والعكس
صحيح

4) The marginal p.d.f of Y is

cont

$$f_y(y) = \sum_{x\in R_x} f(x,y)$$

$$f_y(y) = \int_{R_x} f(x,y)\,dx.$$

اذا بدي الـ marginal
p.d.f بالنسبة للـ Y
بكامل على X والعكس

smile for life

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لما يطلب اشي بدلالة الـ X
ممكن نستعمل marginal p.d.f
للـ X واذا طلب للـ Y نفس الاشي

No.

5) $EX = \sum_{x \in R_x} x f_x(x)$ → Cont. $EX = \int_{R_x} x f_x(x)\, dx$

- $EX^2 = \sum_{x \in R_x} x^2 f_x(x)$ $\int_{R_x} x^2 f_x(x)\, dx$

* $Eg(x) = \int_{R_x} g(x) f_x(x)\, (dx)$

6) $EY = \sum_{R_y} y f_y(y)$ cont. → $\int_{R_y} y f_y(y)\, dy$

- $EY^2 = \sum_{y \in R_y} y^2 f_y(y)$ → $= \int_{R_y} y^2 f_y(y)\, dy$

7) $Exy = \sum_{x \in R_x} \sum_{y \in R_y} xy f(x,y)$

لو بدي توقع X و Y
ما بقدر استعمل
X sub X
A sub Y
لنه التوقع مكس

cont:

$$Exy = \iint_{R_{xy}} xy f(x,y)\, dx\, dy$$

- $Eg(x,y) = \iint g(x,y) f(x,y)\, (dx\, dy)$

independent Y و X اذا كان
وطلعنا
marginal p.d.f x
marginal p.d.f y
وضربناهم ببعض
بعطيني f(x,y)
سواء كان cont or Discrete

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8) If X,y are indep. then

$$F(x,y) = P\{X=x, Y=y\}$$

joint د

حاصل ضرب د

marginal.

$$= P\{X=x\} \cdot P(Y=y)$$

$$= f_x(x) \cdot f_y(y).$$

9) If x,y are indep. then

$$EXy = \sum_{x \in R_x} \sum_{y \in R_y} xy\, f(x,y) = \sum_{x \in R_x} \sum_{y \in R_y} xy\, f_x(x) f_y(y)$$

$$= \sum_{x \in R_x} x f_x(x) \sum_{y \in R_y} y f_x(y)$$

$$= (Ex)(Ey)$$

اذا كان **X**و**Y**
Ex الـ **independent**
بتوزع على **X** و**Y** دايما الـ
Co-Var

يساوي 0 لما يكون **X** و **Y**
indep فقط!!

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لنه لما يكونو معتمدين بكم

Monday
23/10

No.

هاض القانون بتطبق على الحالتين وممكن يكون
Sigma in Disc
Integral in Cont.

10) $Var\,X = EX^2 - (EX)^2$
$Var\,Y = EY^2 - (EY)^2$

توقع الثابت بعطيني نفسه

11) $E(aX+b) = aEX + b$
$E(cY+d) = cEY + d$

12) The Co-Variance of X, Y is given by

التباين المشترك بجمع **X** و **Y**

$Cov(X,Y) = EXY - (EX)(EY)$

↳ **Notes:**

a) $Cov(X,X) = EXX - (EX)(EX)$
$= EX^2 - (EX)^2 = Var(X)$

b) $Cov(X,Y)$ could be (-ve)

c) $Cov(aX+b, cY+d)$ لحطه
$= (a)(c)\,Cov(X,Y)$

d) If X, Y are indep then, **Co-0**
$Cov(X,Y) = EXY - (EX)(EY)$ **Var**
$= (EX)(EY) - (EX)(EY) = 0$

13) The correlation coefficient: ← بيرسون معامل
of X, Y is given by

$Corr(X,Y) = \rho(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$; $\sigma_X = \sqrt{Var}$; $\sigma_Y = \sqrt{Var}$

44

Independent = Uncorrelated
dependent = Correlation

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#Hint
Correlation = Corr

*Notes: (about correlation coefficient):

اذا كانت X وY موجب 1 بتكون علاقةطرديه
واذا كان سالب بتكون علاقة قوية عكسية
وكل ما اقتربت من ال0 بتضعف العلاقة

a) $-1 \leq corr(x,y) \leq 1$

b) $corr(aX+b, cY+d)$

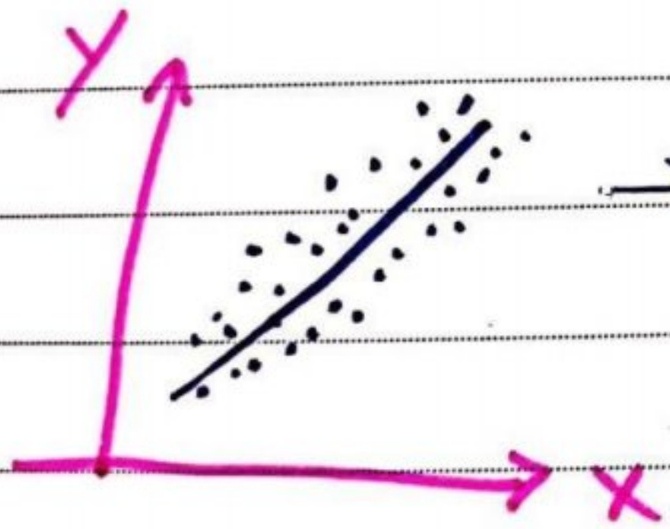
$$= \begin{cases} corr(x,y) & \text{if } a,c \text{ have same sign} \\ -corr(x,y) & \text{if } a,c \text{ have diff. Signs.} \end{cases}$$

اذا كانو معاملات X , Y نفس
الإشارة بنحط
موجب واذا عكس بنحط سالب

c) If x, y are indep. then

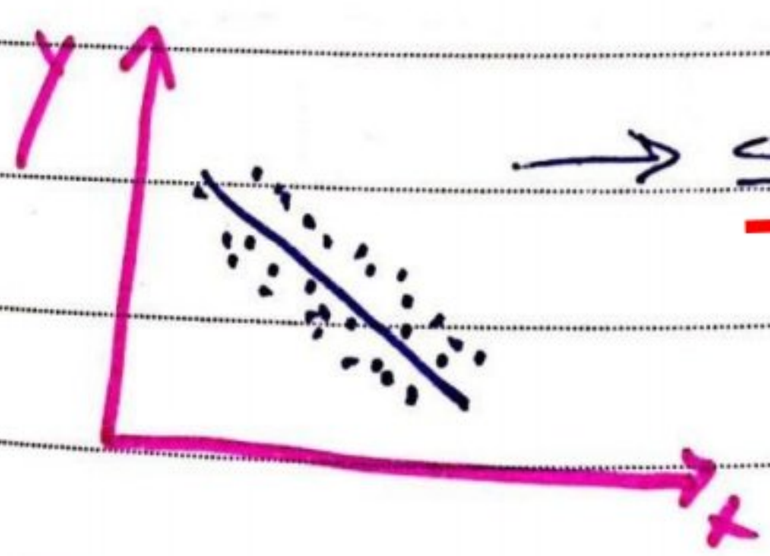
$$corr(x,y) = \frac{cov(x,y)}{\sigma x \sigma y} = 0$$

d) If $corr(x,y) \approx 1$, then



→ strong relation between x, y
(+ve direction) علاقة طردية

e) If $corr(x,y) \approx -1$, then



→ strong relation
(-ve direction) علاقة عكسية

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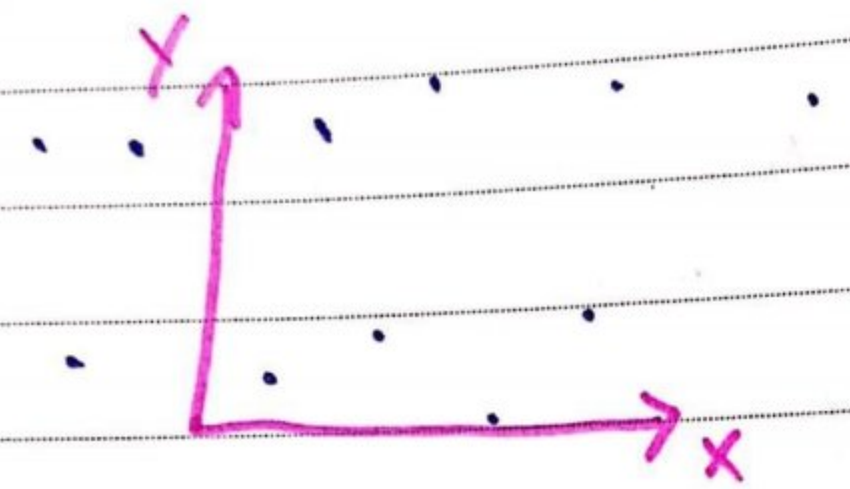
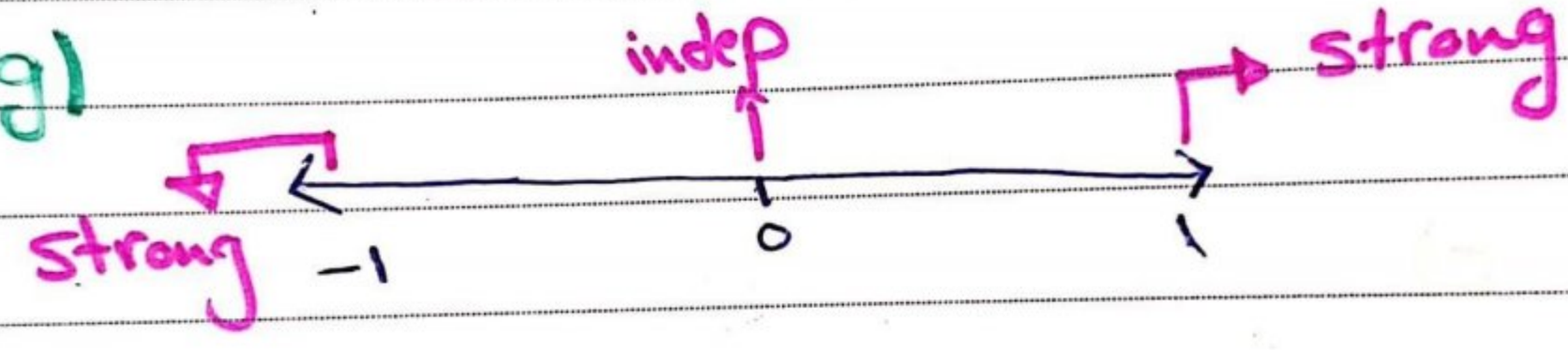
f) If corr(x,y) = 0, then



No relation
(indep)

g)



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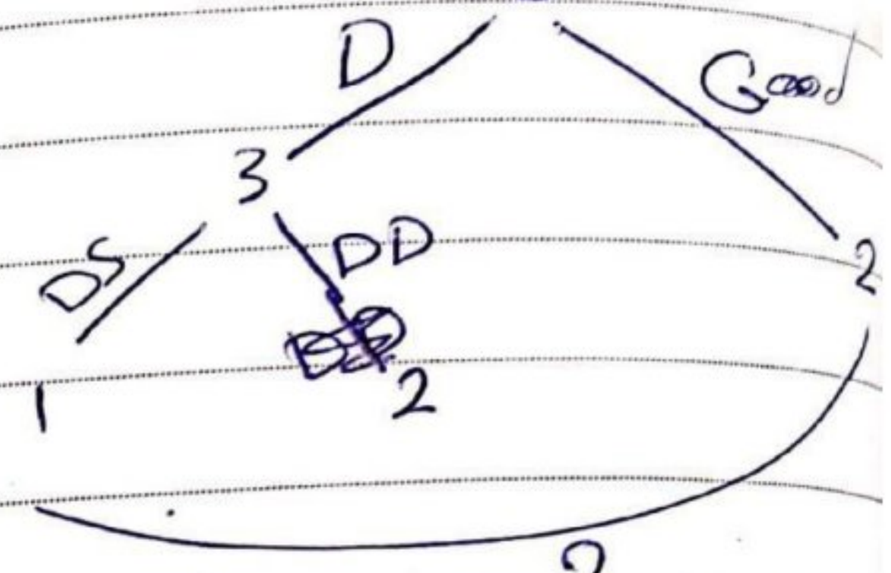
No.

Ex) Three of ~~five~~ items are defective 8-t
1 item is defective in size and 2 items
are defective in design. (DD) (Ds)
Two items are drawn randomly.

let X: no. of Ds items
Y: no. of DD items.

Find every thing:



$X = \{0, 1\}$
$Y = \{0, 1, 2\}$
$R_{X,Y} = \{(0,0), (0,1), (0,2), (1,0), (1,1), (1,2)\}$

Joint p.d.f جدول

Y \ X	0	1	Σ
0	0.1	0.2	0.3
1	0.4	0.2	0.6
2	0.1	0	0.1
Σ	0.6	0.4 →	1

$P(0,0) = \frac{\binom{1}{0}\binom{2}{0}\binom{2}{2}}{\binom{5}{2}} =$

$P(1,0) = \frac{\binom{1}{1}\binom{2}{0}\binom{2}{1}}{\binom{5}{2}} =$

$P(0,1) = \frac{\binom{1}{0}\binom{2}{1}\binom{2}{1}}{\binom{5}{2}} =$

$f(1,1) = \frac{\binom{1}{1}\binom{2}{1}\binom{2}{0}}{\binom{5}{2}} =$

$P(0,2) = \frac{\binom{1}{0}\binom{2}{2}\binom{2}{0}}{\binom{5}{2}} =$

$P(1,2) = 0$

2-way table
- joint p.d.f

مجموعهم كلهم ١
حتى يكون p.d.f

لو ما كان المجموع ١ يكون في خطأ بالحل

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NO.

Wensday
25/10

The marginal p.d.f of X is:

X	$f_X(x)$	$x f_X(x)$	$x^2 f_X(x)$
0	0.6	0	0
1	0.4	0.4	0.4
	1	0.4	0.4

$\rightsquigarrow EX = \sum_{x \in R_X} x f_X(x) = 0.4$

$\rightsquigarrow EX^2 = \sum_{x \in R_X} x^2 f_X(x) = 0.4$

$\rightsquigarrow E(3-2x) = 3 - 2EX = 3 - (2)(0.4) = 2.2$

$\rightsquigarrow E(2x^2 + 5x - 15) = 2EX^2 + 5EX - 15 = (2)(0.4) + (5)(0.4) - 15 =$

$\rightsquigarrow VarX = EX^2 - (EX)^2 = 0.4 - (0.4)^2 = 0.24$

$\rightsquigarrow Var(2-3x) = (-3)^2 VarX = (9)(0.24) = --$

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y	$f_Y(y)$	$y f_Y(y)$	$y^2 f_Y(y)$
0	0.3	0	0
1	0.6	0.6	0.6
2	0.1	0.2	0.4
Σ	1	0.8	1

$$EY = \sum_{y \in R_Y} y f_Y(y) = 0.8$$

$$EY^2 = \sum_{y \in R_Y} y^2 f_Y(y) = 1$$

$$\text{Var}Y = EY^2 - (EY^2) = 1 - (0.8)^2 = 0.36$$

$$EXY = \sum_{x \in R_X} \sum_{y \in R_Y} xy f(x,y)$$

$$= (0)(0) f(0,0) + (0)(1) f(0,1) + (0)(2) f(0,2) + (1)(0) f(1,0) + (1)(1) f(1,1) + (1)(2) f(1,2)$$

$$= 0.2$$

Are X, Y indep.?

$$EXY \stackrel{?}{=} (EX)(EY)$$

$$0.2 \neq (0.4)(0.8)$$

So, X, Y are not indep.

مهم

لو كانت المسلواة متحققه بكونو indep

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(c)

$$\text{Cov}(X,Y) = EXY - (EX)(EY)$$
$$= 0.2 - (0.4)(0.8) = -0.12$$

$$\text{cov}(2X-3, 3-4Y)$$
$$= (2)(-4)\ \text{cov}(X,Y)$$
$$= (-8)(-0.12) = 0.96$$

$$\text{Corr}(X,Y) = \frac{\text{Cov}(X,Y)}{\sigma_X\ \sigma_Y} = \frac{-0.12}{\sqrt{0.24}\sqrt{0.36}} = -0.41$$

$$\text{Corr}(2X-3, 3-4Y) = -\text{Corr}(X,Y) = 0.41$$

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Monday
30/10/17

* Joint distribution function :- (Joint cdf) :

في حالة Cont. ا. فقط

.. The joint cdf or simply the joint d.f. is given by

$$F_{XY}(x,y) = P\{X \le x , Y \le y\}$$

and can be calculated as:

$$F_{XY}(x,y) = \int_{-\infty}^{y}\int_{-\infty}^{x} f(u,v)\, du\, dv$$

تعويض مباشر

$$\int_0^1 e^{x^2} dx \quad \leftarrow \text{لا يمكن حلها}$$

$$\int_0^1\int_0^x e^{x^2} dy\, dx$$

$$\int_0^1 y e^{x^2} \Big]_0^x dx$$

$$\int_0^1 x e^{x^2} dx$$

• Properties :-

1) $0 \le F_{XY}(x,y) \le 1$

2) $F_{XY}(-\infty,-\infty) = 0$ ← اول قيمة بالفترة

$F_{XY}(\infty,\infty) = 1$ ← اخر قيمة بالفترة

3) $F_{XY}(-\infty,y) = 0$, $F_{XY}(x,-\infty) = 0$

لو كان في سالب ما لانهاية بتصفر

4) $f(x,y) = \dfrac{\partial^2}{\partial x\, \partial y} F(x,y)$

5) $f_X(x) = \int_{-\infty}^{\infty} f(x,y)\, dy$

$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\, dx$

مهم جدا

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6) $F_X(x) = \int_{-\infty}^{x} f_X(u)\,du$

- $F_Y(y) = \int_{-\infty}^{y} f_Y(v)\,dv$

7) $F_X(x) = \int_{-\infty}^{x}\int_{-\infty}^{\infty} f(u,y)\,dy\,du$ → marginal p.d.f

مشطوب

- $F_Y(y) = \int_{-\infty}^{y}\int_{-\infty}^{\infty} f(x,v)\,dx\,dv$

8) $f_X(x) = \frac{d}{dx} F_X(x)$

$f_Y(y) = \frac{d}{dy} F_Y(y)$

9) $P(X \in A, Y \in B) = \iint_{B}\int_{A} f(x,y)\,dx\,dy$ هام جدا

$P(x_1 \le X \le x_2, y_1 \le y \le y_2) = \int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x,y)\,dx\,dy$

10) $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

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Ex: $\int_0^1 \left(\int_0^2 xy \, dy \right) dx$ "$x \in [0,1]$ & $y \in [0,2]$"

$= \int_0^1 x \left. \frac{y^2}{2} \right|_0^2 dx$

$= \int_0^1 x \left(\frac{4}{2} - 0 \right) dx$

$= 2 \int_0^1 x \, dx = \left. x^2 \right|_0^1 = 1^2 - 0^2 = 1$

$\int_0^1 \int_0^2 xy \, dy \, dx = \int_0^2 \int_0^1 xy \, dx \, dy$ note

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Ex) let

$$f(x,y) = \begin{cases} b e^{-x} \cos y, & 0 \le x \le 2,\ 0 \le y \le \pi/2 \\ 0, & \text{O.W} \end{cases}$$

a) Find b b) Find $F_{xy}(x,y)$

c) Find $f_x(x)$, $f_y(y)$

d) Find $F_x(x)$, $F_y(y)$

e) Find $P\{0 \le x \le 1,\ 0 \le y \le \pi/4\}$

f) ~~Find $P\{0 \le x \le 1,\ y = \pi/4\}$~~

Sol:

a) $\int_0^{\pi/2}\int_0^{2} b e^{-x} \cos y \, dx\, dy = 1$

$\int_0^{\pi/2} \left(-b e^{-x} \cos y\right]_0^{2} dy = 1$

$\int_0^{\pi/2} \left(-b \cos y (e^{-2} - e^{0})\right) dy = 1$

$-b \sin y (e^{-2} - 1) \Big|_0^{\pi/2} = 1$

$-b(1)(e^{-2} - 1) - 0 = 1 \rightarrow b = \frac{-1}{e^{-2} - 1}$

$= \frac{1}{1 - e^{-2}} = \frac{1}{1 - \frac{1}{e^2}} = \frac{1}{\frac{e^2}{e^2} - \frac{1}{e^2}} = \boxed{\frac{e^2}{e^2 - 1}}$

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No.

$$f(x,y) = \frac{e^2}{e^2-1} e^{-x} \cos y, \quad 0 \le x \le 2, \quad 0 \le y \le \pi/2$$

b) $$F_{XY}(x,y) = \int_0^y \int_0^x \frac{e^2}{e^2-1} e^{-u} \cos v \, du \, dv$$

$$= \frac{e^2}{e^2-1} \int_0^y -e^{-u} \cos v \Big]_0^x dv$$

$$= \frac{e^2}{e^2-1} \int_0^y \cos v \, (1 - e^{-x}) \, dv$$

$$= \frac{e^2}{e^2-1} (1 - e^{-x}) \sin v \Big]_0^y = \frac{e^2}{e^2-1} (1 - e^{-x}) \sin y$$

c) $$f_X(x) = \int_0^{\pi/2} \frac{e^2}{e^2-1} e^{-x} \cos y \, dy$$

$$= \frac{e^2}{e^2-1} e^{-x} \sin y \Big]_0^{\pi/2}$$

$$= \frac{e^2}{e^2-1} e^{-x}$$

$$f_Y(y) = \int_0^2 \frac{e^2}{e^2-1} e^{-x} \cos y \, dx$$

$$= \frac{-e^2}{e^2-1} \cos y \, e^{-x} \Big]_0^2 = \frac{-e^2}{e^2-1} \cdot \cos y \, (e^{-2} - 1)$$

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No.

d) $F_X(x) = \int_0^x f_X(u)\,du$

$$= \frac{e^2}{e^2-1}\int_0^x e^{-u}\,du$$

$$= \frac{-e^2}{e^2-1}\, e^{-u}\Big]_0^x$$

$$= \frac{-e^2}{e^2-1}\left(e^{-x}-1\right)$$

$$F_Y(y) = \int_0^y \frac{-e^2}{e^2-1}\left(e^{-2}-1\right)\cos v\,dv$$

$$= \frac{-e^2}{e^2-1}\left(e^{-2}-1\right)\sin v\Big]_0^y$$

$$= \frac{-e^2}{e^2-1}\left(e^{-2}-1\right)\sin y$$

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Unit step function

$$u(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$u(x - x_0) = \begin{cases} 1, & x \geq x_0 \\ 0, & x < x_0 \end{cases}$$

Note:

$$f(x)\,u(x) = \begin{cases} f(x), & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$f(x)\,u(x-1) = \begin{cases} f(x), & x \geq 1 \\ 0, & x < 1 \end{cases}$$

Impulse fn.

$\delta(x)$ satisfies the impulse fn.

$$\int_{-\infty}^{\infty} \varphi(x)\,\delta(x - x_0)\,dx = \varphi(x_0)$$

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Ex. $\int_{-\infty}^{\infty} x^2 \delta(x-2)\,dx = x^2\Big|_{x=2} = 4$

$\int_{-\infty}^{\infty} \delta(x)\,dx = 1$

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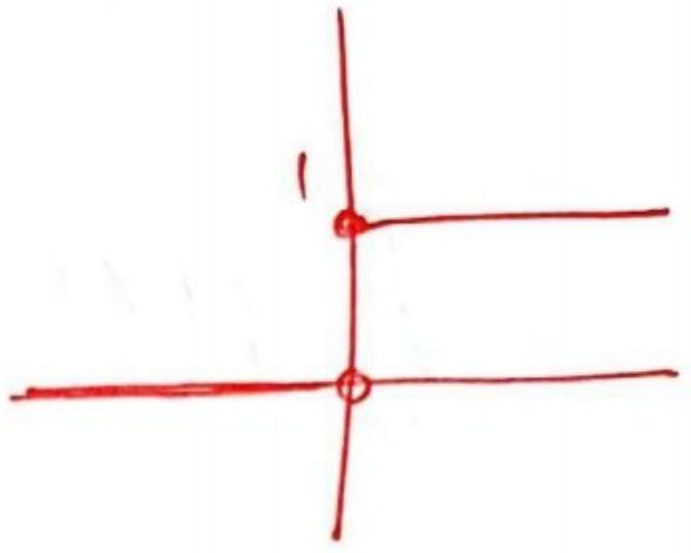
unit step fn.

$$u(x) = \begin{cases} 1 , & x \geq 0 \\ 0 , & x < 0 \end{cases}$$

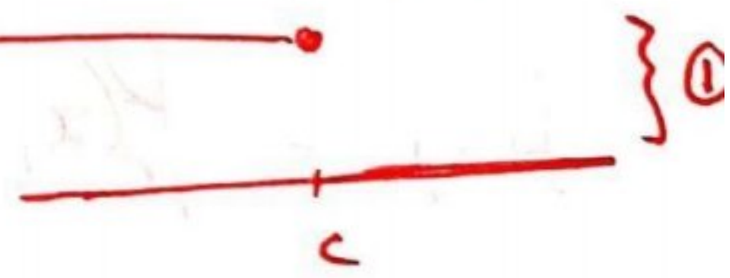


$$f(x)\,u(x) = \begin{cases} f(x), & x \geq 0 \\ 0 , & x < 0 \end{cases}$$

$$u(x-c) = \begin{cases} 1 , & x \geq c \\ 0 , & x < c \end{cases}$$

$$f(x)\,u(x-c) = \begin{cases} f(x) , & x \geq c \\ 0 , & x < c \end{cases}$$

$$u(c-x) = \begin{cases} 0 , & x > c \\ 1 , & x \leq c \end{cases}$$



هوم نكم أس نيبطخ نيبطخ ا ب...

1. Find the generalize p.d.f
2. Find the p.d.f imbalance

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unit impulse function

$$\int_{-\infty}^{\infty} \varphi(x)\,\delta(x-c)\,dx = \varphi(c)$$

$$\int_{-\infty}^{\infty} \delta(x-c)\,dx = 1$$

Note: $\dfrac{du(x)}{dx} = \delta(x)$

Note:- Let $x \in \{x_1, x_2, \cdots, x_n\}$ be

discrete r.v.

→ The p.d.f

$$f(x) = p_r(x_1)\,\delta(x-x_1) + p_r(x_2)\,\delta(x-x_2) + \cdots + p_r(x_n)\,\delta(x-x_n)$$

& the c.d.f

$$F(x) = p_r(x_1)\,u(x-x_1) + p_r(x_2)\,u(x-x_2) + \cdots + p_r(x_n)\,u(x-x_n)$$

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الدكتور بده يعيده وبدي اعدله..

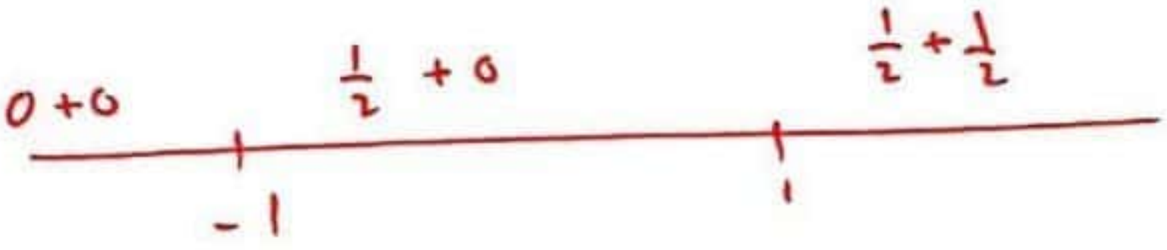
Ex. X discrete r.v.

$S = \{1, -1\}$

$f(1) = \frac{1}{2}$ & $f(-1) = \frac{1}{2}$

$\rightarrow f(x) = \frac{1}{2}\delta(x-1) + \frac{1}{2}\delta(x+1)$

$F(x) = \frac{1}{2}u(x-1) + \frac{1}{2}u(x+1)$



Ex. X cont. r.v. with c.d.f

$F(x) = u(x)\left(1 - e^{-x^2/b}\right)$

. Then

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Ex.

$$f(x,y) = u(x)\,u(y)\,x\,e^{-x(y+1)}$$

find $f_x(x)$, $f_y(y)$, $F_x(x)$, $F_y(y)$

$F_{xy}(x,y)$,

Sol.

$$f_x(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$

$$= \int_{-\infty}^{\infty} u(x)\,u(y)\,x\,e^{-x(y+1)}\,dy$$

$$= \int_{0}^{\infty} u(x)\,x\,e^{-x(y+1)}\,dy$$

$$= u(x)\,x\,e^{-x}\int_{0}^{\infty} e^{-xy}\,dy$$

$$= u(x)\,x\,e^{-x}\left(\frac{1}{-x}e^{-xy}\Big|_0^{\infty}\right)$$

$$= u(x)\,e^{-x}\,(1-0)$$

$$= u(x)\,e^{-x}$$

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$$f_y(y) = \int_{-\infty}^{\infty} u(x)\,u(y)\,x\,e^{-x(y+1)}\,dx$$

$$= \int_{0}^{\infty} u(y)\,x\,e^{-x(y+1)}\,dx$$

$$= u(y)\left[\frac{-x}{y+1}e^{-x(y+1)} - \frac{1}{(y+1)^2}e^{-x(y+1)}\right]_0^{\infty}$$

$$= u(y)\left[0 - 0 - 0 + \frac{1}{(y+1)^2}\right)$$

$$= \frac{u(y)}{(y+1)^2}$$

x		$e^{-x(y+1)}$	
	$+$		$-\frac{1}{y+1}e^{-x(y+1)}$
1		$e^{-x(y+1)}$	
	$-$		$\frac{1}{(y+1)^2}e^{-x(y+1)}$
0			

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No.

- $F_X(x) = \int_0^x f_X(u)\, du$

$= \int u(x)\, e^{-x}\, dx$

$= \int_0^x e^{-u}\, du \longrightarrow -e^{-u}\Big]_0^x = -(e^{-x} - 1) = 1 - e^{-x}$

- $F_Y(y) = \int u(y) \frac{1}{(y+1)^2}\, dy$

$= \int_0^y \frac{1}{(v+1)^2}\, dv = \int_0^y (v+1)^{-2}\, dv$

$= \frac{(v+1)^{-1}}{-1}\Big]_0^y \longrightarrow \frac{(y+1)^{-1}}{-1} - \frac{1}{-1}$

$= 1 - \frac{1}{y+1}$

- $F_{XY}(x,y) = \int_0^y \int_0^x u\, e^{-u(v+1)}\, du\, dv.$ ⋮

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* Prob. Distribution:-

عدد مرات التجربة
بنسميها N
نجاح التجربة P
فشل التجربة Q

* Discrete Dist. :-

- Binomial dist. :-
- ~~Bois~~ Binomial experement:-

خواص التجربة حتى تكون Binomial :

① n indep. and identical trails.

n ← من المحاولات المستقلة المتماثلة

② X: no. of successes out of n trails.

$X = \{0, 1, 2, \ldots, n\} = R_X$

← Range ال تكون محدد من الأدنى لرقم الأعلى

Ex

X no. of heads in 5 trails

$X = \{0, 1, 2, 3, 4, 5\}$

e.g: Y: no. of student who pass an exam out of

$Y = \{0, 1, \ldots, 10\}$

③ output → Success
→ fail

لها وجهتان فقط

④ $P(\text{success}) = p$ (fixed) الاحتمال كل مرة يكون ثابت

$P(\text{fail}) = q = 1 - p$

⑤ The Binomial r.v X is denoted by

$X \sim B(n, p)$

is ←

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⑥ If $X \sim B(n,p)$ then

اذا اجتمعت صفت الايدنتكل والاندبندنت بنستعمل البايناوميل دستريبيوشن

a) $EX = mean = np$

b) $Var\ X = (n*p*q)$

c) $p(X=x) = \binom{n}{x} p^x q^{n-x}$, $x = 0, 1, \dots n$

Ex) In a multiple Choice exam with 10 question, suppose that each question has 5 answer with only one true answer. If you answer all question with pure guess

a) Find the prob. that you will answer only 5 true

b) Find the prob. that you will get a full mark.

c) Find the prob. that you will answer at most 2 questions correctly.

d) Find the mean of true answered questions.

e) Find the variance of true answered questions

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No.

Sol:

let X: no. of true answered question.

$X = \{0, 1, 2, \dots, 10\}$

$n = 10 \qquad p = \frac{1}{5} = 0.2$

$X \sim B(10, 0.2)$

دائما في Bio الاكس تبدأ من 0

a. $p(X=5) = \binom{10}{5} (0.2)^5 (0.8)^{10-5} = 0.026$

b. $p(X=10) = \binom{10}{10} (0.2)^{10} (0.8)^0 = 1.02 \times 10^{-7} = 0.0000001$

c. $p(X \leq 2) = p(X=0) + p(X=1) + p(X=2)$

$= \binom{10}{0} (0.2)^0 (0.8)^{10} + \binom{10}{1} (0.2)^1 (0.8)^9 + \binom{10}{2} (0.2)^2 (0.8)^8 =$

$=$

d. $EX = np$

$= (10)(0.2) = 2$ ← الوسط الحسابي لعدد الاجابات الصحيحة

e. $Var X = n.p.q \qquad = (10)(0.2)(0.8) = 1.6$

f. $std(x) = \sqrt{1.6} = 1.26$

دائما في Bio الاكس تبدأ من 0

Bio Dist الـ فيها 3 براميتر
-N
-P
-Q

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اذا كان في وقت بالسؤال
بكون Poisson dist

بواسون

* Poisson dist :-

- Poisson experiment :-

1) X : No. of occurences per unit of time.

e.g : X : No. of patients entering the ER per hour

$X = \{0, 1, 2, \dots\}$

↳ محدد من الاسفل وغير محدد من الأعلى .

↳ محدد في وقت معين

2) m : average of occurences per unit of time.

3) The poisson r.v. X denoted by

$X \sim p(m)$

4) If $X \sim p(m)$, then

↳ a) $p(X=x) = \frac{e^{-m} m^x}{x!}$, $x = 0, 1, 2, \dots$

↳ b) $EX = \text{Var } X = m$

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No. 13/11

Assume that Average no. of patients entering the emergency room is 1.2 per hour.

a) Find the prob. that only 2 patients will enter per hour.

b) Find the prob. that at least 2 patients will enter per hour.

c) Find the prob. that no. of patients is between 2 and 3 per hour (included) ← بتشمل included القيمة المقفلة

d) Find $P(1.1 < X < 2.9)$

e) Find $P(X = 2.5)$

f) Find EX, $Var\,X$

g) Find the prob. that only 3 patients will enter in 4 hours.

h) Find the average no. of patients that will enter in a day.

Sol:

$X \sim P(1.2)$

a) $P(X=2) = \frac{e^{-1.2}(1.2)^2}{2!} = \dots$

b) $P(X \geq 2) = 1 - P(X < 2)$

$= 1 - [P(X=0) + P(X=1)]$

$= 1 - \left[\frac{e^{-1.2}(1.2)^0}{0!} + \frac{e^{-1.2}(1.2)^1}{1!}\right] = \dots$

اذا كان في وقت بالسؤال بكوّن Poisson dist

كم برامتر بكون في Poisson dist

1 الي هو الافرج اما الاكس لا تعتبر برامتر

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24
12
48
240
28.8

c) $P(2 \le X \le 3) = P(X=2) + P(X=3)$

$= \frac{e^{-1.2}(1.2)^2}{2!} + \frac{e^{-1.2}(1.2)^3}{3!} = \cdots$

d) $P(1.1 < X < 2.9) = P(X=2) = \frac{e^{-1.2}(1.2)^2}{2!} = \cdots$

e) $P(X=2.5) =$ Zero

f) $EX = \text{Var}\, X = 1.2$

g) let $Z = 4X$ (2.)

$\rightarrow Z \sim P((1.2)(4))$

$Z \sim P(4.8)$

$P(Z=3) = \frac{e^{-4.8}(4.8)^3}{3!} = \cdots$

G & H cancellations

h) let $Y = 24X$

$\rightarrow Y \sim P((24)(1.2))$

$Y \sim P(28.8)$

$\rightarrow EY = 28.8$ patients per day

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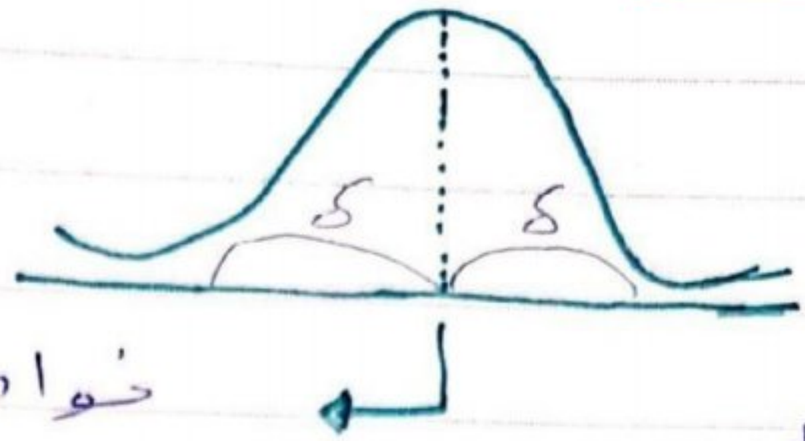
* Normal distribution :-

(Gaussian dist)

(Belt - shaped dist)

متماثل يمين ويسار



خواص هذه النقطة : يتساوى الوسط مع المنوال مع

1) mean = EX = median = mode = μ

median ← القيمة التي تتساوى فيها الأعلى مع الأدنى

له القيمة الأكثر تكراراً .

الرسمة بتكون متماثلة يمين ويسار

2) symmetric dist. نقطة التماثل

3) The normal r.v X depends only on μ and σ^2. So, we denoted it by:

$$X \sim N(\mu, \sigma^2)$$

μ ← الوسط الحسابي ، σ^2 ← Var

4) If $X \sim N(\mu, \sigma^2)$. then

a) $EX = \mu$, $\text{Var } X = \sigma^2$

p.d.f ←

b) $$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} \quad , -\infty < x < \infty$$

c) $P(a \le x \le b) = P(a < x < b)$

$$= \int_a^b f(x)\,dx = \text{area under } f(x) \text{ between a and b}$$

$$= F(b) - F(a)$$

$$= \int_{-\infty}^{b} f(x)\,dx - \int_{-\infty}^{a} f(x)\,dx$$

= area to the left of b − area to the left of a

ال cdf لعدد هي المساحة على يسار العدد

smile for life

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1.

d) $P(X=c) = 0$

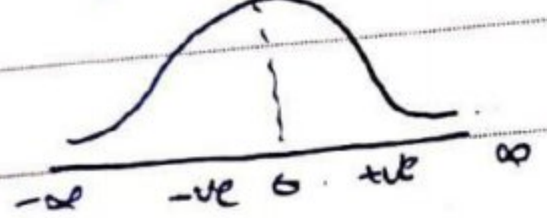
e) $P(X \leq b) = \int_{-\infty}^{b} f(x)\,dx = F(b)$ = area to the left of

* Standard normal distribution:-

let $X \sim N(\mu, \sigma^2)$, then the r.v.

$Z = \frac{X-\mu}{\sigma} \sim N(0,1) \equiv$ standard Normal

Properties:-

1) 

2) Symmetric about 0.

3) $EZ = 0$, $Var\ Z = 1$

4) $f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$, $-\infty < z < \infty$

5) $P(a \leq z \leq b) = \int_{a}^{b} f(z)\,dz$

= area under $f(z)$ between a and

= $T(b) - T(a)$

= table at b - table at a

= area to the left of b - area to the left of a

ال cdf بشكل عام
F الا بالستاندر
بتكون T

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6) $\int_{-\infty}^{\infty} f(z)\,dz = 1$ = total area

7) $\int_{-\infty}^{0} f(z)\,dz = 0.5 = \int_{0}^{\infty} f(z)\,dz$ لذة متماثل حول النقطة.

8) $p(z \le b) = T(b)$

9) $p(z > b) = 1 - p(z < b) = 1 - T(b)$

10) $p(z = b) = 0$

دايما 0 لنه كونتينيوس راندم فاريبل

لو بدي المساحة على يمين عدد بطلع المساحة على اليسار وبطرحها من 1

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No.

15/11/2017

Ex) Find $p(Z<1.15) = T(1.15) = 0.87493$

Ex) $p(-1.25 < Z < 2.15)$

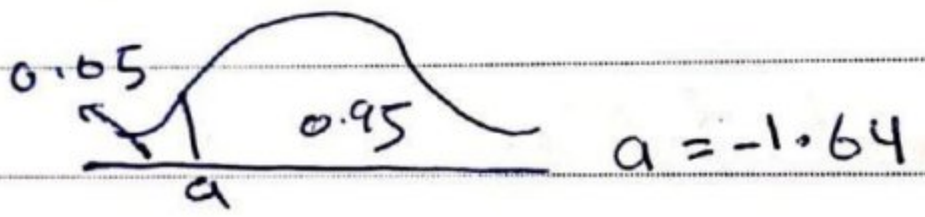
$= T(2.15) - T(-1.25)$

$= 0.98422 - 0.10565$

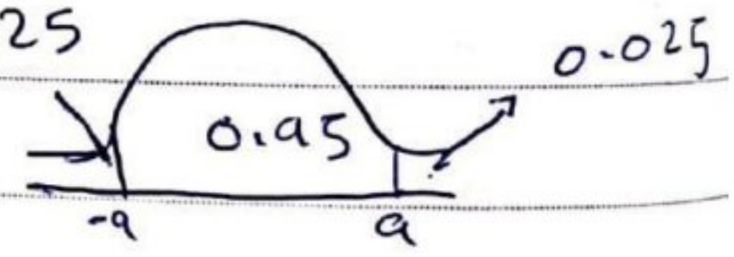
Ex) Find a s.t. $p(Z<a) = 0.9$

Ex) Find a s.t

$p(Z>a) = 0.95$



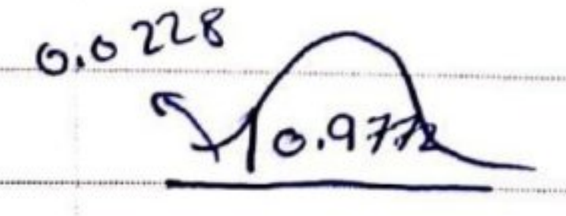
Ex) Find a s.t. $p(-a<Z<a) = 0.95$



we want a s.t $p(Z<a) = 0.975$

$a = 1.96$

Ex) Find a s.t $p(Z>a) = 0.9772$



$a = -2$

$1 - 0.9772$

$= 0.0228$

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Ex) Suppose that the ranges of some signals are normally distributed with mean 15 Kms (μ) and std 3 Kms (σ).

15~ 19

3^2 ←

a) Find the proportion of the signals that exceeds 17 Kms.

[If a signal is chosen at random, find the prob. that its range will exceed 17 Kms

b) Find the proportion of signals with ranges less than 10 Kms

c) Find the proportion of signals with ranges between 10 and 17 Kms.

d) Find the range of signal that are best than 90% of other signals.

e) Find the range of signal that is considred with max range of the worst 5% of signals.

f) $P(X=16)$

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No.

8

g) If two signals are chosen at random, find the prob. that their ranges will exceed 17 Kms.

h) If 5 signals are chosen at random, find the prob. that the ranges of two of them will exceed 17 Kms.

$$Z=(X-\mu)/\sigma$$

Sol:-

$$X \sim N(15, 9)$$

a) $P(X>17) = P\left(\frac{X-\mu}{\sigma} > \frac{17-\mu}{\sigma}\right)$

$= P\left(Z > \frac{17-15}{3}\right)$ $= P(Z > 0.67)$

$= 1 - P(Z \leq 0.67)$

$= 1 - T(0.67)$

$21 - 0.74857$

20.25143

b) $P(X<10) = P\left(\frac{X-\mu}{\sigma} < \frac{10-\mu}{\sigma}\right)$

$= P\left(Z < \frac{10-15}{3}\right)$

$= P(Z < -1.67)$

$= T(-1.67)$ $= 0.04746$

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c) $P(10 < X < 17)$

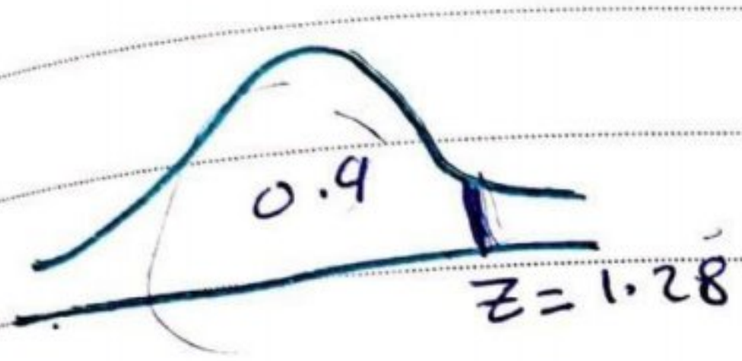
$= P\left(\frac{10-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{17-\mu}{\sigma}\right)$

$= P(-1.67 < Z < 0.67)$

$= T(0.67) - T(-1.67)$

$= 0.74857 - 0.04746 = \ldots\ldots$

d)

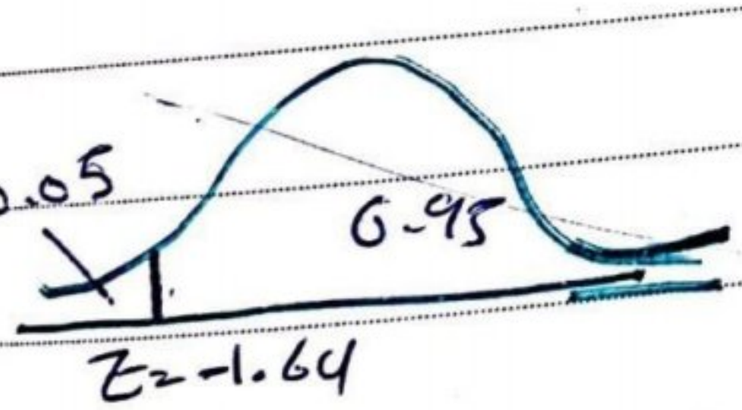


but $Z = \frac{X-\mu}{\sigma}$

$1.28 = \frac{X-15}{3} \rightarrow X = (3)(1.28) + 15$

$= 18.84$ Kms

e)



$X = \sigma Z + \mu$

$= (3)(-1.64) + 15 = 10.08$ Kms

f) $P(X=16) = 0$

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(15)

g) $p(X_1 > 17) = 0.25143$ (من الفرع a)

$p(X_2 > 17) = 0.25143$

هون مختار 2 سقنل وبده 2 يكونو صح

we want

$\rightarrow p(X_1 > 17 \cap X_2 > 17)$

$= p(X_1 > 17) \cdot p(X_2 > 17)$

$= (0.25143)^2$

h) $p(X_1 > 17) = 0.25143$

$p(X_2 > 17) = 0.25143$

هون مختار 5 سقنل وبده 2 يكونو صح عشان هيك حله بختلف عن الي فوق لو طلب 5 صح بنحله زي الي فوق

$\vdots$

$p(X_5 > 17) = 0.25143$

All signals are indep. and identical $n=5$

$\rightarrow Y \sim B(5, 0.25143)$

$\leftarrow P(y=2) = \binom{5}{2} (0.25143)^2 (1-0.25143)^{5-2}$ (binomial)

$= \dots$

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STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409

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0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
3.0	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997

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Transformation of random variable:-

If $f_X(x)$ is the p.d.f of a r.v. X and $\boxed{Y = T(X)}$ is a function of the r.v. X, we want to find the distribution of a r.v. Y

$$F_Y(y_0) = P\{Y \le y_0\}$$

ال3 اسطر اثبات مش معنا

$$= P\{T(X) \le y_0\}$$

$$= P\{X \le T^{-1}(y_0)\} = F_X(x_0)$$

Hence,

$$f_Y(y) = f_X(T^{-1}(y)) \left| \frac{d\,T^{-1}(y)}{dy} \right| \rightarrow \text{Jacobian}$$

ex) Let $Y = aX + b$, then

$$X = \frac{Y-b}{a} \longrightarrow \frac{dx}{dy} = \frac{1}{a}$$

$$f_Y(y) = f_X\left(\frac{Y-b}{a}\right) \left|\frac{1}{a}\right|$$

ex) If $Y = aX^2 + b$, then

$$X = \pm\sqrt{\frac{Y-b}{a}} \longrightarrow \frac{dx}{dy} = \frac{\frac{1}{a}}{\pm 2\sqrt{\frac{Y-b}{a}}}$$

$$f_Y(y) = F_X\left(\pm\sqrt{\frac{Y-b}{a}}\right) \left| \frac{\frac{1}{a}}{\pm 2\sqrt{\frac{Y-b}{a}}} \right|$$

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Ex] Consider the dist. with p.d.f

$$f(x) = \frac{\theta^2}{\theta+1}(1+x)e^{-\theta x}, \quad x>0$$

Find the density of Y s.t.

$$Y = e^X - 1, \quad y>0.$$

بنخلي x موضع قانون

$$y+1 = e^x \longrightarrow x = \ln(y+1)$$

$$\frac{dx}{dy} = \frac{1}{y+1}$$

$$f_Y(y) = f_X(\ln(y+1)) \left|\frac{1}{y+1}\right|$$

$$= \frac{\theta^2}{\theta+1}\left(1+\ln(y+1)\right) e^{-\theta(\ln(y+1))} \left|\frac{1}{y+1}\right|$$

$$= \frac{\theta^2}{\theta+1}\left(1+\ln(y+1)\right)(y+1)^{-\theta}\frac{1}{y+1}$$

$$= \frac{\theta^2}{\theta+1}\left(1+\ln(y+1)\right)\frac{1}{(y+1)^{\theta+1}}$$

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$$y = \frac{1}{x} \longrightarrow x = \frac{1}{y} \longrightarrow \frac{dx}{dy} = -\frac{1}{y^2}$$

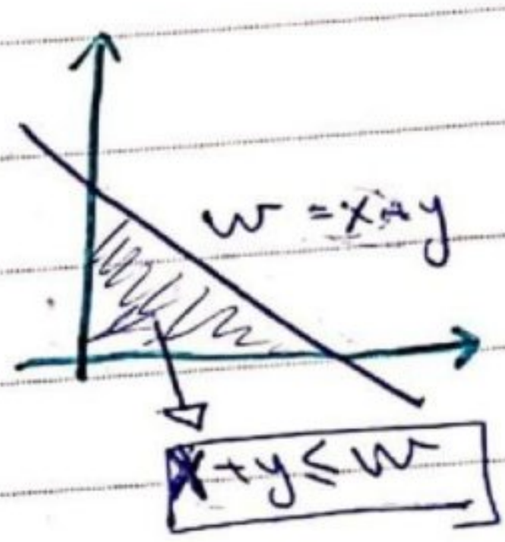
$$f_Y(y) = f_X\left(\frac{1}{y}\right)\left|-\frac{1}{y^2}\right|$$

(*) Distribution of sum of two random variables:-

Let X, Y be two indep. r.v's and $\boxed{W = X+Y}$

$$F_W(w) = P\{W \le w\} = P\{X+Y \le w\}$$

هون بدنا نوجد pdf للـ w



$$= \int_{-\infty}^{\infty}\int_{-\infty}^{w-y} f(x,y)\,dx\,dy$$

$$f_W(w) = \int_{-\infty}^{\infty} f_Y(y) \int_{-\infty}^{w-y} f_X(x)\,dx\,dy$$

by diff.

$$f_W(w) = \int_{-\infty}^{\infty} f_Y(y)\, f_X(w-y)\,dy .$$

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No.

Ex] Find the density of $W = X + Y$ where

$$f_X(x) = \frac{1}{a}\left[u(x) - u(x-a)\right]$$

$$f_Y(y) = \frac{1}{b}\left[u(y) - u(y-b)\right]$$

$x = w - y$ نعوض بدل x

يو اف واي ناقص عدد هو واحد على يمين العدد وصفر على يساره

$$* f_W(w) = \int_{-\infty}^{\infty} \frac{1}{b}\left[u(y) - u(y-b)\right] \cdot \frac{1}{a}\left[u(w-y) - u(w-y-a)\right]$$

$$= \frac{1}{ab}\int_{0}^{\infty}\left[1 - u(y-b)\right]\left[u(w-y) - u(w-y-a)\right] dy$$

ضرب ← القوسين

يو اف عدد ناقص واي صفر على يمين العدد وواحد على يسارة

تكامل ← طرح

$$= \frac{1}{ab}\int_{0}^{\infty} \underbrace{\left(u(w-y)\right)}_{①} - \underbrace{u(w-y-a)}_{②} - \underbrace{u(y-b)\,u(w-y)}_{③} + \underbrace{u(y-b)\,u(w-y-a)}_{④}\, dy$$

الـ A < B والـ2 موجبات

$$① = \frac{1}{ab}\int_{0}^{\infty} u(w-y)\,dy = \frac{1}{ab}\int_{0}^{w} 1_o\, dy = \frac{w}{ab}$$

$$② = \frac{-1}{ab}\int_{0}^{\infty} u(w-y-a)\,dy = \frac{-1}{ab}\int_{0}^{w-a} 1 \cdot dy = -\left(\frac{w-a}{ab}\right)$$

IF W>=A

If w-a >= 0 , and 0 o.w

$$③ = \frac{-1}{ab}\int_{0}^{\infty} u(y-b)\,u(w-y)\,dy = \frac{1}{ab}\int_{b}^{w} 1 \cdot dy = -\left(\frac{w-b}{ab}\right)$$

: ؛ w ، b

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④ $= \frac{1}{ab} \int_0^{\infty} u(y-b)\, u(w-a-y)\, dy$

$= \frac{1}{ab} \int_b^{w-a} 1\, dy \quad = \frac{w-a-b}{ab}$

If w-a >= b , and 0 o.w

- $f_W(w) = \begin{cases} ① & , \ w < a \\ ①+② & , \ a \le w < b \\ ①+②+③ & , \ b \le w < a+b \\ ①+②+③+④ & , \ w \ge a+b \end{cases}$

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$$\textcircled{1} = \frac{\omega}{ab}$$

$$\textcircled{2} = 0$$

$$\textcircled{3} = 0$$

$$\textcircled{4} = 0$$

$$\begin{cases} \textcircled{1} = \dfrac{\omega}{ab} \\ \textcircled{2} = -\dfrac{(\omega-a)}{ab} \\ \textcircled{3} = 0 \\ \textcircled{4} = 0 \end{cases}$$

$$\begin{cases} \textcircled{1} = \dfrac{\omega}{ab} \\ \textcircled{2} = -\left(\dfrac{\omega-a}{ab}\right) \\ \textcircled{3} = -\left(\dfrac{\omega-b}{ab}\right) \\ \textcircled{4} = 0 \end{cases}$$

$$\begin{cases} \textcircled{1} = \dfrac{\omega}{ab} \\ \textcircled{2} = \dfrac{-(\omega-a)}{ab} \\ \textcircled{3} = -\left(\dfrac{\omega-b}{ab}\right) \\ \textcircled{4} = \dfrac{\omega-a-b}{ab} \end{cases}$$

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$$f_W(w) = \int_{-\infty}^{\infty} f_y(y)\, f_x(w-y)\, dy$$

Ex) $f_x(x) = \frac{1}{a}[u(x) - u(x-a)]$

$$f_y(y) = b\, u(y)\, e^{-by}$$

Find the density of $W = X+Y$
s.t. X, Y are indep.

$$f_W(w) = \int_{-\infty}^{\infty} f_y(y)\, f_x(w-y)\, dy$$

$$= \int_{-\infty}^{\infty} b\, u(y)\, e^{-by} \frac{1}{a}[u(w-y) - u(w-y-a)]\, dy$$

$$= \frac{b}{a}\int_{0}^{\infty} e^{-by}[u(w-y) - u(w-a-y)]\, dy$$

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No.

$$= \frac{b}{a} \int_0^{\infty} \left(\underbrace{e^{-by}\, u(w-y)}_{①} - \underbrace{e^{-by}\, u(w-a-y)}_{②} \right) dy$$

$$① = \frac{b}{a} \int_0^{\infty} e^{-by}\, u(w-y)\, dy$$

$$= \frac{b}{a} \int_0^{w} e^{-by} \cdot 1\, dy = \frac{b}{a} \cdot \frac{-1}{b} e^{-by} \Big]_0^{w}$$

$$= \frac{-1}{a} \left(e^{-bw} - 1 \right)$$

$$② = \frac{-b}{a} \int_0^{\infty} e^{-by}\, u(w-a-y)\, dy$$

$$= \frac{-b}{a} \int_0^{w-a} e^{-by}\, dy = \frac{-b}{a} \cdot \frac{-1}{b} e^{-by} \Big]_0^{w-a}$$

$$= \frac{1}{a} \left(e^{-b(w-a)} - 1 \right)$$

$$f_W(w) = \begin{cases} ① & , \ w < a \\ ① + ② & , \ a \leq w \end{cases}$$

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No.

⊗ Expected value of functions of r.v's :-

- Recall; $$EX = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

$$EY = \int_{-\infty}^{\infty} y f_Y(y)\, dy$$

$$EXY = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x y\, f(x,y)\, dx\, dy$$

$$Eg(x) = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

$$Eg(y) = \int_{-\infty}^{\infty} g(y) f_Y(y)\, dy$$

$$Eg(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y) f(x,y)\, dx\, dy$$

* Special Case :-

⇝ The joint moment about the origin is given by:

$$m_{nk} = E(X^n Y^k) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x^n y^k f(x,y)\, dx\, dy$$

$$m_{n0} = E(X^n) = \int_{-\infty}^{\infty} x^n f_X(x)\, dx$$

is called the n^{th} moment of X

$$m_{0k} = E(Y^k) = \int_{-\infty}^{\infty} y^k f_Y(y)\, dy$$

is called the k^{th} moment of Y

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⊗ Expected value of functions of r.v's :-

- Recall; $EX = \int_{-\infty}^{\infty} x f_X(x)\, dx$

$$EY = \int_{-\infty}^{\infty} y f_Y(y)\, dy$$

$$EXY = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f(x,y)\, dx\, dy$$

$$Eg(x) = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

$$Eg(y) = \int_{-\infty}^{\infty} g(y) f_Y(y)\, dy$$

$$Eg(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y) f(x,y)\, dx\, dy$$

* Special case :-

↝ The joint moment about the origin is given by:

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$$m_{0k} = E(Y^k) = \int_{-\infty}^{\infty} y^k f_Y(y)\, dy$$

is called the K^{th} moment of Y

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The 1^{st} moment of X is EX_1 = mean = μ,

The 2^{nd} moment of X is EX^2

$\text{Var } X = EX^2 - (EX)^2$

$= 2^{nd} \text{ moment} - (1^{st} \text{ moment})^2$

* $n+K$ is called the order of the moment.

* $m_{11} = E(XY)$ is called the correlation.

$$m_{11} = E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f(x,y)\, dx\, dy$$

= Correlation of X_1Y

$= R_{XY}$.

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(*) If X, y are indep., then

$$EXy = (EX)(EY)$$

$$m_{11} = (m_{10})(m_{01}).$$

* If $R_{xy} = 0 = Exy$,
then X, y are orthogonal.

Ex) let X be a r.v. with $EX = 3$, $6x^2 = 2$
and $Y = -6X + 22$.

a) Find the correlation of X, y
b) Are X, y correlated.

sol:

$$m_{11} = EXY$$

$$= E(X(-6X + 22))$$

$$= E(-6X^2 + 22X)$$

$$= -6EX^2 + 22EX$$

But $6x^2 = EX^2 - (EX)^2$

$$2 = EX^2 - (3)^2 \qquad EX^2 = 11$$

So, $EXy = -6EX^2 + 22EX$

$= (-6)(111 + (22)(3)) = 0$ مهم

So, X, y are orthogonal

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b) $EXY \overset{?}{=} (EX)(EY)$

Now, $EY = E(-6X + 22)$

$= -6EX + 22$

$= (-6)(3) + 22 = 4$

So, $EXY \neq (EX)(EY)$

$0 \neq (3)(4)$

So, X, Y are not indep.
X, Y are depent.
X, Y are correlated.

* If $EXY = (EX)(EY)$, then
X, Y are indep.
X, Y are uncorrelated

* If $Y = aX + b$, $a \neq 0$, then
X, Y are correlated.

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⊛ Joint Central Moment :-

It is defined as:-

$$\mu_{nk} = E\left((x-\bar{x})^n (y-\bar{y})^k\right)$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x-\bar{x})^n (y-\bar{y})^k f(x,y)\, dx\, dy$$

Notice that:

$$\mu_{20} = E(x-\bar{x})^2 = \int_{-\infty}^{\infty} (x-\bar{x})^2 f_x(x)\, dx = \sigma_x^2$$

$$\mu_{02} = E(y-\bar{y})^2 = \sigma_x^2$$

and

$$\mu_{11} = E\left((x-\bar{x})(y-\bar{y})\right)$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x-\bar{x})(y-\bar{y}) f(x,y)\, dx\, dy$$

$$= C_{xy} = Cov(x,y)$$

= is the covariance of x, y

$$= Exy - (Ex)(Ey)$$

$$= Rxy - (m_{10})(m_{01})$$

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~ If X, Y are orthogonal, then

$$C_{XY} = \text{Cov}(X,Y) = -(EX)(EY)$$

~ If X, Y are indep. (uncorrelation) then

$$C_{XY} = \text{Cov}(X,Y) = 0$$

~ Correlation Coefficient

$$= \text{Corr}(X,Y) = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{C_{XY}}{\sigma_X \sigma_Y}$$

$$= \frac{0}{\sigma_X \sigma_Y} = 0$$

* The Uniform distribution:-

The Uniform r.v. is defined as

$X \sim U(a,b)$ with p.d.f.

$$f(x) = \begin{cases} \frac{1}{b-a} & , a \le x \le b \\ 0 & , \text{o.w} \end{cases}$$

and c.d.f.

$$F(x) = \begin{cases} 0 & , x < a \\ \frac{x-a}{b-a} & , a \le x < b \\ 1 & , x \ge b \end{cases}$$

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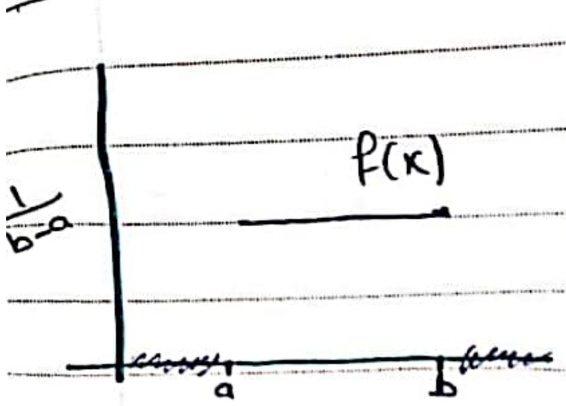

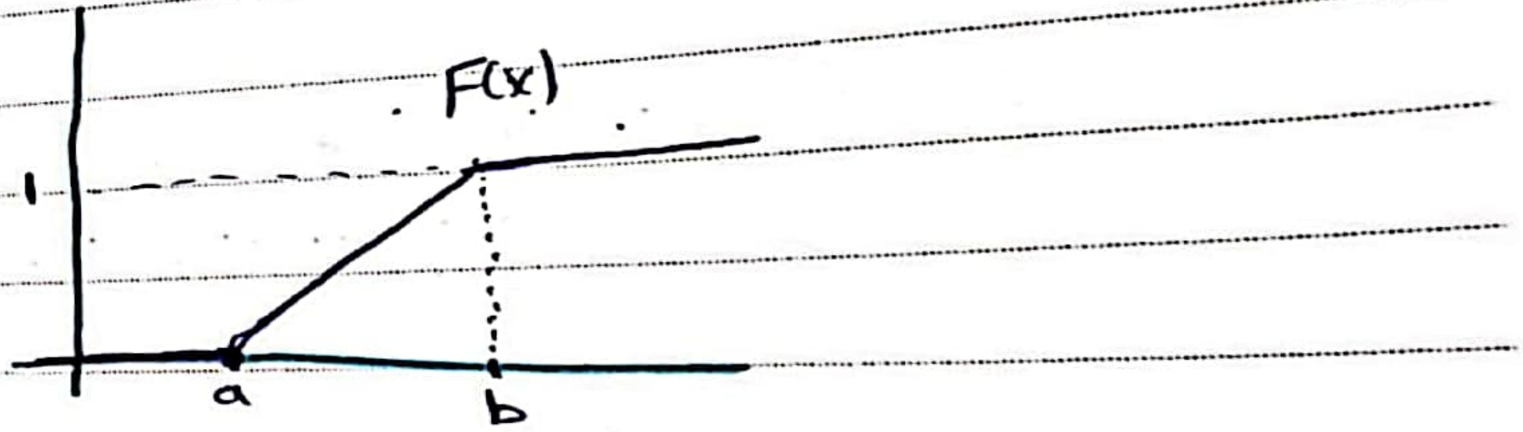

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Random process

Def A random process (RP) [or stochastic process] is an infinite indexed family $X(t,s) = \{x(t,s)\}$

"$s \in \mathcal{S}$ is a possible outcome of an experiment"

note $x(t,s_0)$ (s_0 fixed) is called sample function or ensemble function or realization of a process.

note $x(t_0,s)$ (t_0 fixed) is a random variable on $\mathcal{S}$

note For frozen time $t = t_1$

$$X_1 := X(t_1,s) = X(t_1)$$

describes statistically the properties of the R.V at time t_1

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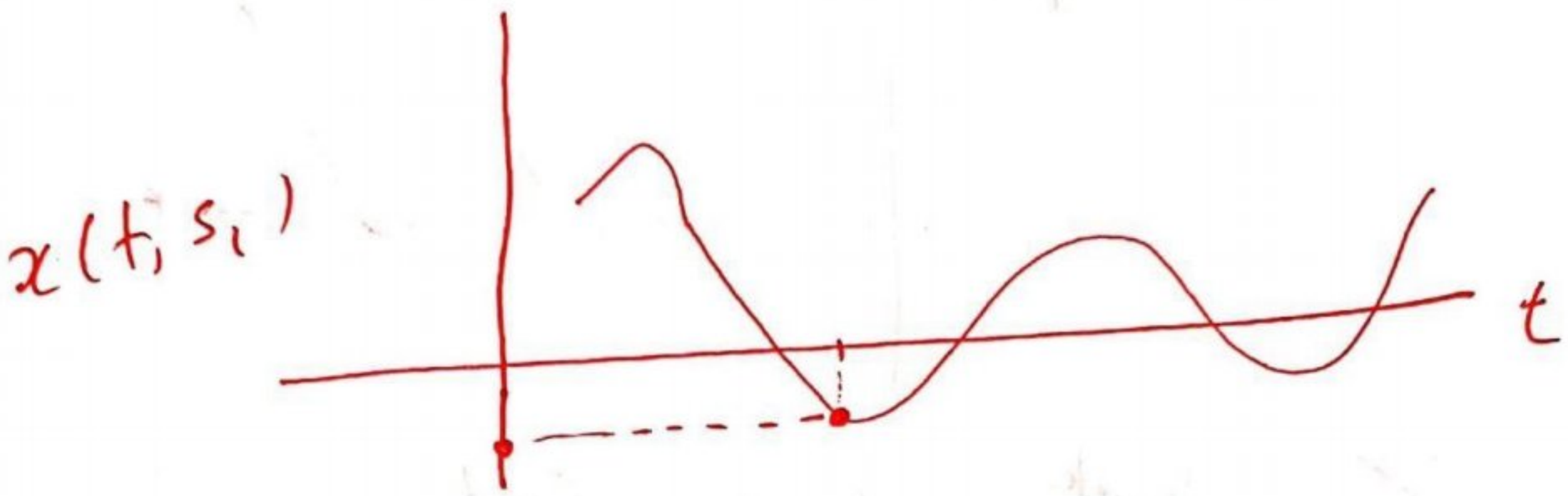
$E(X_1)$ = mean value of the process

is called ensemble average



note

In general

$X_i = X(t_i, s) = X(t_i)$

$i = 1, 2, \ldots$

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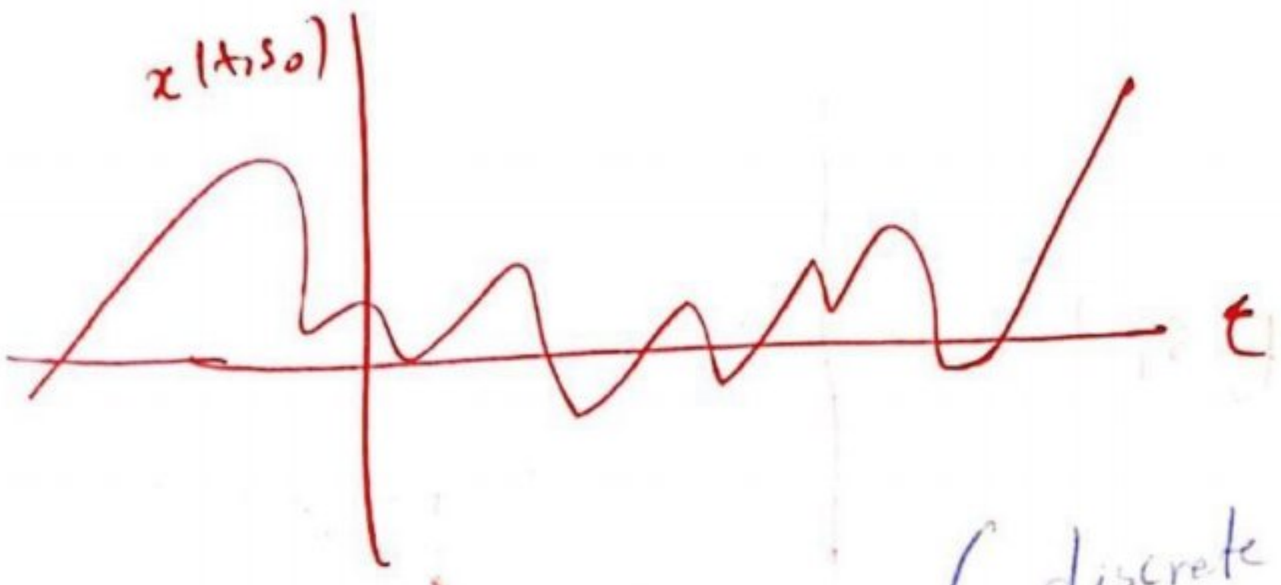
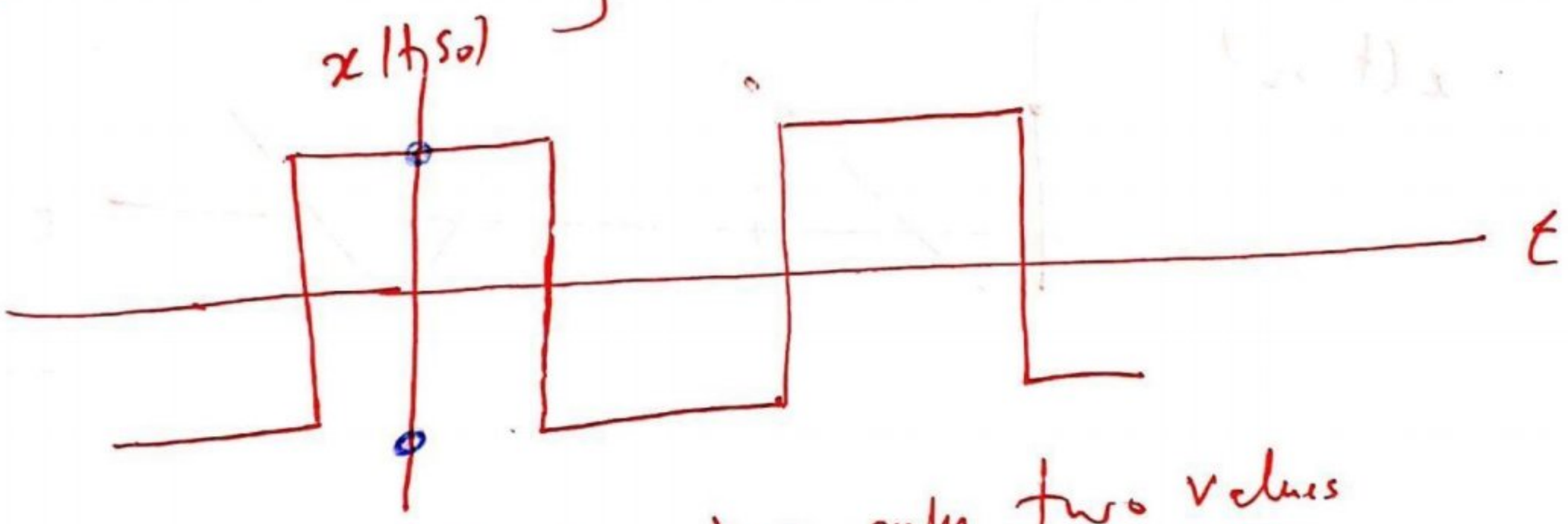
classification of process

① X is cont & t is cont

X(t) is called Continuous r.p



② X is discrete & t is cont. (discrete random process)

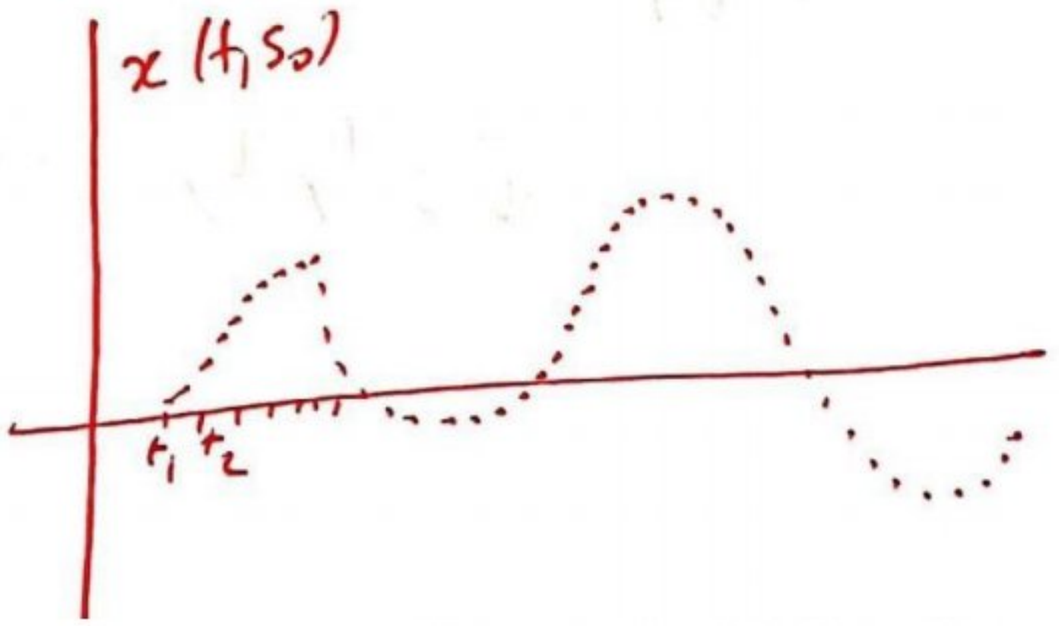


The sample function has only two values

③ X is cont, & t is discrete

(Cont. random sequence)

(discrete time "DT" R.P)



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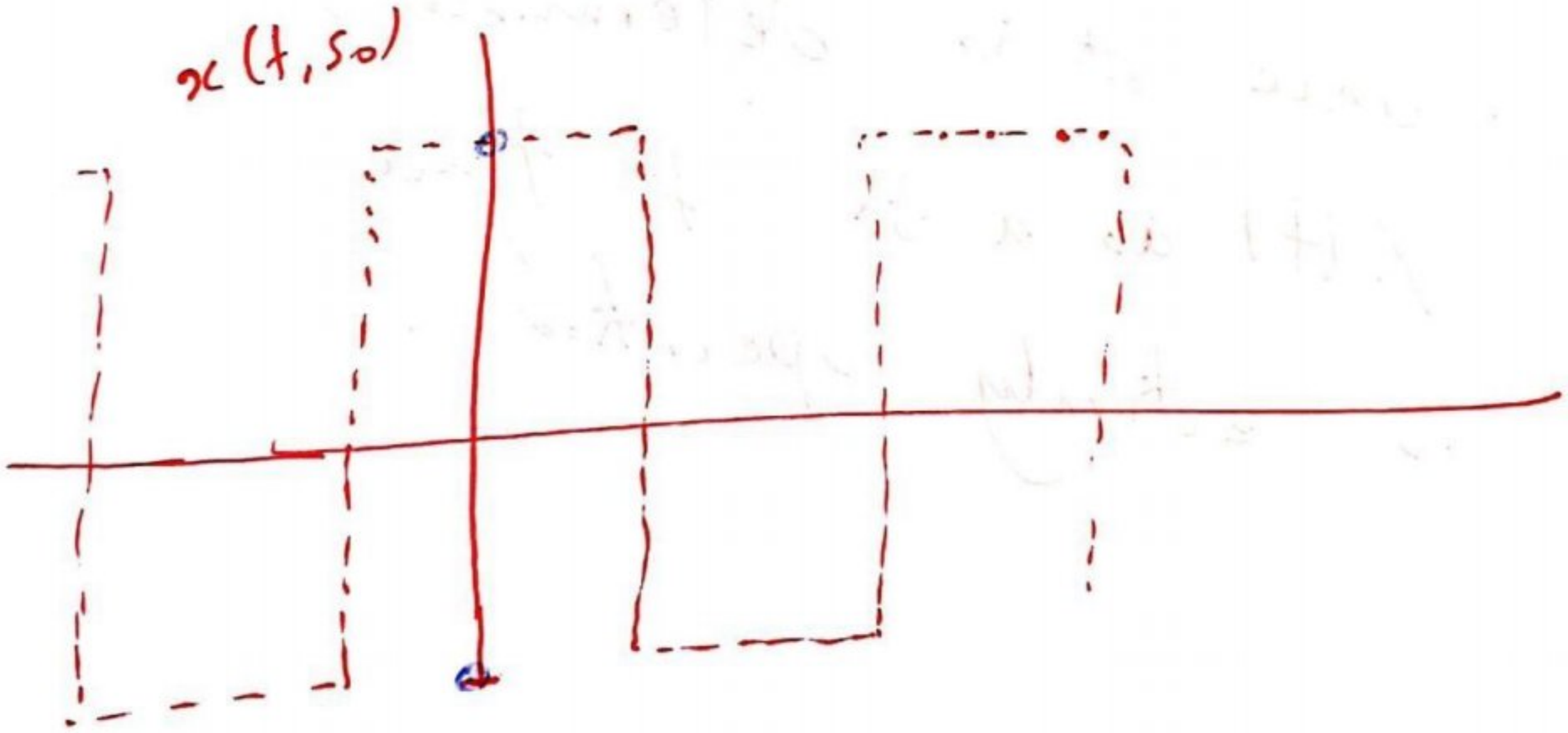
A DT random process is a set of random variables $\{X(t_n)\}$, for sample times $t_n = nT_s$, $n = 0, \pm 1, \pm 2, \ldots$

T_s := sampling interval

$1/T_s$ = sampling rate "sample per second"

④ X & t discrete
(discrete r.p)



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Def In a deterministic R.P / any future values of any sample fn can be predicted from the past ones :

ex : $X(t) = A \cos(\omega t + \theta)$

where A & ω are fixed

θ r.v

θ is determined,

note : Once θ is ... $X(t)$ as a fn of time is entirely specified.

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No.

Def: The random process is called stationary if all of its statistical properties do not change with time.

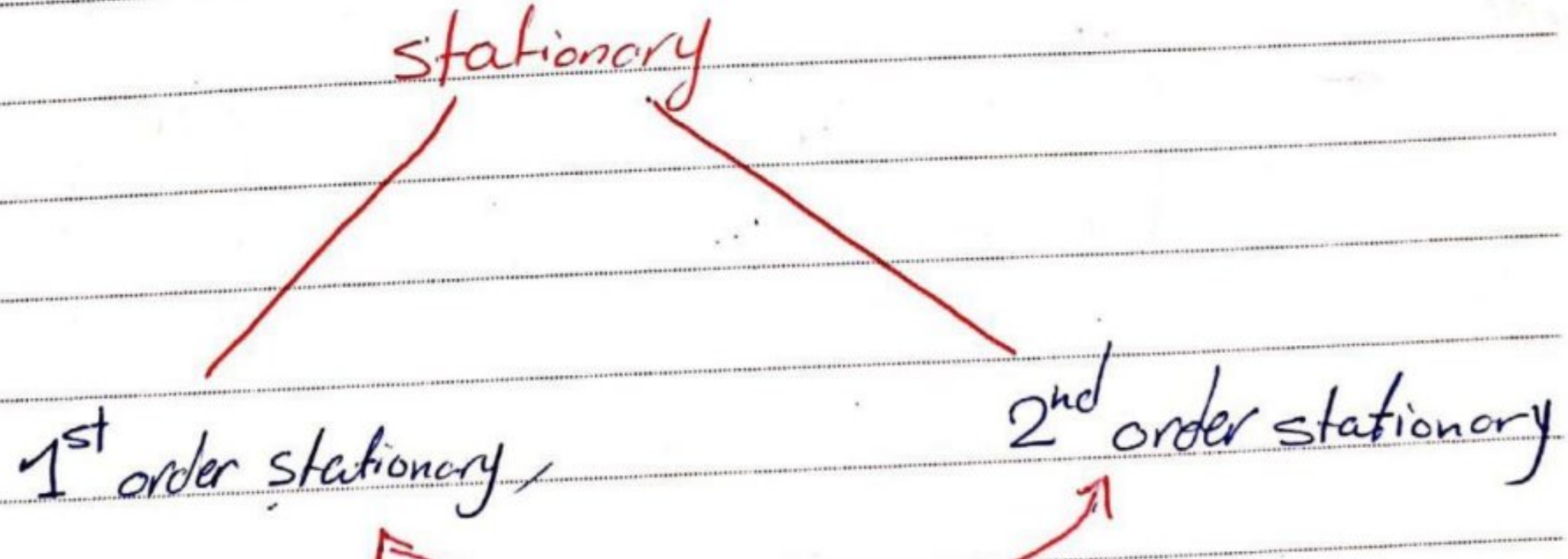
i.e. At time t_1, the c.d.f. of the random variable $X_1 = x(t_1)$

is

$$F_X(x_1, t_1) = P\{X(t_1) \leq x_1\}, \quad \forall x_1$$

And, for two random variables $X_1 = x(t_1)$, $X_2 = x(t_2)$, the joint c.d.f.

$$F(x_1, x_2, t_1, t_2) = P\{X(t_1) \leq x_1, X(t_2) \leq x_2\}, \quad \forall x_1, x_2$$



بكون wide
لما يتحققو ال2

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Remember that:-

- x, y indep. if $f(x,y) = f_x(x) \cdot f_y(y)$

→ $EXY = EX\,EY$

$Cov(x,y) = C_{xy} = 0$

$Corr(x,y) = 0$

⑤ Independence:-

Two random processes $x(t)$, $y(t)$ are statistically indep. if

$$f(x_1, x_2 \dots x_n, y_1, y_2, \dots, y_m, t_1, t_2, \dots, t_n, t_1', t_2', \dots, t_m')$$

$$= f_X(x_1, x_2 \dots x_n, t_1, t_2, \dots t_n) \cdot f_Y(y_1, y_2, \dots y_m, t_1', t_2'$$

$$\left(f(\tilde{x}, \tilde{y}, \tilde{t}, \tilde{t}') = f_x(\tilde{x}, \tilde{t}) \cdot f_y(\tilde{y}, \tilde{t}')\right.$$

$$\tilde{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \tilde{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}, \tilde{t} = \begin{pmatrix} t_1 \\ \vdots \\ t_n \end{pmatrix}, \tilde{t}' = \begin{pmatrix} \tilde{t_1} \\ \vdots \\ \tilde{t_m} \end{pmatrix}$$

⑥ First order stationary:

A random process is called 1st order-stationary if its first-order density function (p.d.f) do not change with time shift.

i.e: $f_X(x, t_1) = f_X(x, t_1 + \Delta)$

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First order stationary:

A random process is called 1st order-stationary if its first-order density function (p.d.f) do not change with time shift.

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This mean that $f_X(x, t_1)$ is indep. of t_1
and so $E(X(t)) = \overline{X} =$ constant

↳ ليس اقتران بدلالة t
(فيه ثابت)

⊛ Second order stationary :-

A process is called 2nd order stationary if its 2nd order density satisfies

$$f_X(x_1, x_2, t_1, t_2) = f_X(x_1, x_2, t_1 + \Delta, t_2 + \Delta)$$ ⊛

Remarks:

(1) The correlation $E(x_1 x_2) = E(X(t_1)\, X(t_2))$
can be denoted by $R_{XX}(t_1, t_2)$
and it is called the auto-correlation function.
of the random process $X(t)$.

Hense $R_{XX}(t_1, t_2) = E(X(t_1)\, X(t_2))$
let $t_1 = t$, $t_2 = t_1 + \tau$ $(\tau = t_2 - t_1)$

In case of 2nd order stationary, the random process satisfies ⊛ Consequently

$$R_{XX}(t, t+\tau) = E(X(t_1)\, X(t_2 + \tau))$$
$$= E(X(t)\, X(t+\tau))$$
$$= R_{XX}(\tau)$$ Auto-correlation

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* $x(t)$ is called 2^{nd} order stationary if the auto-correlation of $x(t)$

$$R_{xx}(t, t+\tau) = R_{xx}(\tau)$$

The random process is called wide-sense stationary if $E(x(t)) = \bar{x} = \text{Constant}$

$$R_{xx}(t, t+\tau) = R_{xx}(\tau)$$

* Ex] Show that the process $x(t) = A\cos(\omega_0 t + \Theta)$ is wide-sense stationary if A, ω_0 are constants and θ is Uniformly distributed r.v. on $(0, 2\pi)$

Now,

$$E(x(t)) = \int_{-\infty}^{\infty} x(t)\, f(\theta)\, d\theta$$

$$= \int_0^{2\pi} A\cos(\omega_0 t + \Theta) \cdot \frac{1}{2\pi}\, d\theta$$

$$= \frac{A}{2\pi} \int_0^{2\pi} \cos(\omega_0 t + \Theta)\, d\theta$$

$= 0 \equiv$ Constant.

$\rightarrow$ $x(t)$ is 1^{st} order stationary

Remember that

If $X \sim U(a, b)$

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{o.w} \end{cases}$$

$$F(x) = \begin{cases} \frac{x-a}{b-a}, & a \le x < b \\ 0, & \text{o.w} \end{cases}$$

$\Theta \sim U(0, 2\pi)$

$$f(\theta) = \begin{cases} \frac{1}{2\pi - 0}, & 0 \le \theta \le 2\pi \\ 0, & \text{o.w.} \end{cases}$$

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Now, the auto-correlation with

$t_1 = t$, $t_2 = t + \tau$ is

$$R_{xx}(t, t+\tau) = E(x(t) * x(t+\tau))$$

$$= E(A\cos(\omega_0 t + \theta) \cdot A\cos(\omega_0 t + \omega_0 \tau + \theta))$$

$$= A^2 E(\cos(\omega_0 t + \theta) \cdot \cos(\omega_0 t + \omega_0 \tau + \theta))$$

$$= \frac{A^2}{2} E(\cos(2\omega_0 t + \omega_0 \tau + 2\theta) + \cos(\omega_0 \tau))$$

$$= \frac{A^2}{2} E(\cos(2\omega_0 t + \omega_0 \tau + 2\theta)) + \frac{A^2}{2} E(\cos(\omega_0 \tau)) \rightarrow \text{free of } \theta$$

$$\frac{A^2}{2} \int_0^{2\pi} \cos(2\omega_0 t + \omega_0 \tau + 2\theta) \cdot \frac{1}{2\pi} \, d\theta + \frac{A^2}{2} \cos(\omega_0 \tau)$$

$$= \frac{A^2}{2} \cos(\omega_0 \tau) = R_{xx}(\tau)$$

So, $x(t)$ is 2nd order stationary.

$\rightarrow$ $x(t)$ is wide-sense stationary.

Remember that:-

- $\cos(a+b) = \cos a \cos b - \sin a \sin b$
- $\cos(a-b) = \cos a \cos b + \sin a \sin b$
- $\frac{\cos(a+b) + \cos(a-b)}{2} = \cos a \cos b$
- $\frac{\cos(a-b) - \cos(a+b)}{2} = \sin a \sin b$
- $\cos(-x) = \cos x$

  $\sin(-x) = -\sin x$

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NO.

* If we have two random processes $x(t)$, $y(t)$. then,
they are jointly wide-sense stationary if each one of them is wide-sense stationary and their cross-correlation is a function of $\tau = t_2 - t_1$,

$$\text{i.e. } R_{xy}(t, t+\tau) = E(x(t)\, y(t+\tau))$$
$$= R_{xy}(\tau).$$

* Correlation function:-

* Auto-correlation function:-

The auto-correlation function of $x(t)$ is denoted by

$$R_{xx}(t, t+\tau) = E(x(t)\, x(t+\tau))$$

If $x(t)$ is wide-sense stationary, then

$$R_{xx}(t, t+\tau) = R_{xx}(\tau).$$

(*) For any random process, the auto-correlation function satisfies the following:-

1) $|R_{xx}(\tau)| \leq R_{xx}(0)$.

2) $R_{xx}(-\tau) = R_{xx}(\tau)$

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3) $R_{xx}(\infty) = E(X^2(t))$

مهم 4 + 3

4) $\lim_{\tau \to \infty} R_{xx}(\tau) = \bar{X}^2$

شرط كافي

if $E(x(t)) = \bar{X}$ & $X(t) = \bar{X}$ ثابت

5) If $x(t)$ is periodic, then $R_{xx}(\tau)$ is periodic with the same period.

Def: A process $x(t)$ with constant mean value $\bar{X}$ is called mean-ergodic

i.e. $E(x(t)) = \bar{X} = A(x(t)) = \bar{x}(t)$

$$A(\star) = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} \star \, dt$$

Ex) Given the auto-correlation function for a stationary ergodic process with no periodic component is

$$R_{xx}(\tau) = 25 + \frac{4}{1+6\tau^2}$$

- Find the mean and variance of the process.

Sol:

By (4);

$$\lim_{\tau \to \infty} R_{xx}(\tau) = \lim_{\tau \to \infty} \left(25 + \frac{4}{1+6\tau^2}\right) = 25 = \bar{X}^2 \longrightarrow \bar{X} = \pm 5$$

By (3); $R_{xx}(0) = 25 + \frac{4}{1+6(0)^2} = 29 = EX^2_{(t)}$

$\sigma^2_{x(t)} = EX^2_{(t)} - (EX_{(t)})^2$

$= 29 - (\pm 5)^2 = 4$

smile

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* Cross-correlation function:-

The cross-correlation of $x(t)$, $y(t)$ is given by

$$R_{xy}(t, t+\tau) = E(x(t)\, y(t+\tau))$$

* If $x(t)$, $y(t)$ are jointly wide-sense stationary, then

$$R_{xy}(t, t+\tau) = E(x(t)\, y(t+\tau)) = R_{xy}(\tau)$$

* If $R_{xy}(\tau) = 0$, then $x(t)$, $y(t)$ are orthogonal processes.

* If $x(t)$, $y(t)$ are indep., then

$$R_{xy}(t, t+\tau) = E(x(t))\, E(y(t+\tau))$$

If $x(t)$, $y(t)$ are indep and at least wide-sense stationary, then

$$R_{xy}(\tau) = \bar{x}\,\bar{y} \text{ which is constant.}$$

⊛ properties:

1) $R_{xy}(-\tau) = R_{xy}(\tau)$.

2) $|R_{xy}(\tau)| \leq \sqrt{R_{xx}(0)\, R_{yy}(0)}$

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3) $|R_{xy}(\tau)| \leq \frac{1}{2}[R_{xx}(0) + R_{yy}(0)]$

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Ex Assume the two random process

$$x(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)$$
$$y(t) = B\cos(\omega_0 t) - A\sin(\omega_0 t),$$

where A, B are r.v's, ~~of~~ constant, A, B are uncorrelated, Zero-mean with same variance.

Find the cross-correlation $R_{xy}(t, t+\tau)$ and show that $x(t), y(t)$ are jointly wide-sense stationary.

$$E(A) = E(B) = 0$$
$$E(AB) \overset{\text{indep}}{=} E(A)E(B) = 0$$

$$\sigma_A^2 = \sigma_B^2 = \sigma^2$$

$$R_{xy}(t, t+\tau) = E(x(t)\, y(t+\tau))$$
$$= E(\{A\cos(\omega_0 t) + B\sin(\omega_0 t)\}.$$
$$\{B\cos(\omega_0 t + \omega_0\tau) - A\sin(\omega_0 t + \omega_0\tau)\})$$

$$= E(AB\cos(\omega_0 t)\cos(\omega_0 t + \omega_0\tau) - A^2\cos(\omega_0 t)\sin(\omega_0 t + \omega_0\tau)$$
$$+ B^2\sin(\omega_0 t)\cos(\omega_0 t + \omega_0\tau) - AB\sin(\omega_0 t)$$
$$\sin(\omega_0 t + \omega_0\tau))$$

$$= E(AB)E(\cos(\omega_0 t)\cos(\omega_0 t + \omega_0\tau)) - E(A^2)E(\cos(\omega_0 t)\sin(\omega_0 t$$
$$+ E(B^2)E(\sin(\omega_0 t)\cos(\omega_0 t + \omega_0\tau)) - E(AB)E(\sin(\omega_0 t)\sin($$
$$(\omega_0 t$$

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No.

$$\sigma_A^2 = E(A^2) - (E(A))^2 \to 0$$

$$\sigma_B^2 = E(B^2) - (E(B))^2 \to 0$$

But $\sigma_A^2 = \sigma_B^2 = \sigma^2 = E(A^2) = E(B^2)$

$$= -\sigma^2 E(\cos(\omega_0 t)\sin(\omega_0 t + \omega_0 \tau)) + \sigma^2 E(\sin(\omega_0 t)\cos(\omega_0 t + \omega_0 \tau))$$

$\sin(a-b) = \sin a \cos b - \cos a \sin b$

$$= \sigma^2 E(\sin(\omega_0 t - \omega_0 t - \omega_0 \tau))$$

$$= -\sigma^2 E(\sin(\omega_0 \tau))$$

$$= -\sigma^2 \sin(\omega_0 \tau)$$

$$= R_{xy}(\tau)$$

So, $X(t)$, $Y(t)$ are jointly wide-sense stationary. (#)

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No.

Ex) let $X(t)$ be a wide-sense stationary with auto-correlation $R_{xx}(\tau) = e^{-a|\tau|}$, (وجب a)

let $Y(t) = X(t)\cos(\omega_0 t + \Theta)$, where Θ is uniform r.v. on $(-\pi, \pi)$, ω_0 is constant and Θ is indep. of $X(t)$, find the auto-correlation of $Y(t)$.

$\cos(\omega_0 t + \Theta)$ ←

Sol:- $R_{yy}(t, t+\tau) = E(y(t)\, y(t+\tau))$

$= E(X(t)\cos(\omega_0 t + \Theta)\, X(t+\tau)\cos(\omega_0 t + \omega_0 \tau + \Theta))$

$= E(X(t)X(t+\tau))\, E(\cos(\omega_0 t + \Theta)\cos(\omega_0 t + \omega_0 \tau + \Theta))$

$= R_{xx}(t, t+\tau) \int_{-\pi}^{\pi} \left(\cos\left(\frac{2\omega_0 t + 2\Theta + \omega_0\tau}{2}\right) + \cos(\omega_0\tau)\right) \frac{1}{2\pi}\, d\theta$

$= R_{xx}(\tau) \cdot \frac{1}{2\pi}\left[0 + \cos(\omega_0\tau)(2\pi)\right]$

$= R_{xx}(\tau)\cos(\omega_0\tau)$

$= e^{-a|\tau|}\cos(\omega_0\tau) = R_{yy}(\tau)$.

So, it is wide-sense stationary.

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* Cross-correlation function:-

The cross-correlation of $x(t)$, $y(t)$ is given by

$$R_{xy}(t, t+\tau) = E(x(t)\, y(t+\tau))$$

* If $x(t)$, $y(t)$ are jointly wide-sense stationary, then

$$R_{xy}(t, t+\tau) = E(x(t)\, y(t+\tau)) = R_{xy}(\tau)$$

* If $R_{xy}(\tau) = 0$, then $x(t)$, $y(t)$ are orthogonal processes.

* If $x(t)$, $y(t)$ are indep., then

$$R_{xy}(t, t+\tau) = E(x(t))\, E(y(t+\tau))$$

If $x(t)$, $y(t)$ are indep and at least wide-sense stationary, then

$$R_{xy}(\tau) = \bar{x}\,\bar{y} \text{ which is constant.}$$

Ref page w.s

⊛ properties:

1) $R_{xy}(-\tau) = R_{xy}(\tau)$.

2) $|R_{xy}(\tau)| \leq \sqrt{R_{xx}(0)\, R_{yy}(0)}$

3) $|R_{xy}(\tau)| \leq \frac{1}{2}[R_{xx}(0) + R_{yy}(0)]$

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Ex Assume the two random process

$$x(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)$$
$$y(t) = B\cos(\omega_0 t) - A\sin(\omega_0 t),$$

where A, B are r.v's, ω_0 constant, A, B are uncorrelated, Zero-mean with same variance.

Find the cross-correlation $R_{xy}(t, t+\tau)$ and show that $x(t), y(t)$ are jointly wide-sense stationary.

$$E(A) = \underset{\text{indep}}{E(B)} = 0$$
$$E(AB) = E(A)E(B) = 0$$

$$\sigma_A^2 = \sigma_B^2 = \sigma^2$$

$$R_{xy}(t, t+\tau) = E(x(t)\, y(t+\tau))$$
$$= E(\{A\cos(\omega_0 t) + B\sin(\omega_0 t)\}.$$
$$\{B\cos(\omega_0 t + \omega_0\tau) - A\sin(\omega_0 t + \omega_0\tau)\})$$

$$= E(AB\cos(\omega_0 t)\cos(\omega_0 t + \omega_0\tau) - A^2\cos(\omega_0 t)\sin(\omega_0 t + \omega_0\tau)$$
$$+ B^2\sin(\omega_0 t)\cos(\omega_0 t + \omega_0\tau) - AB\sin(\omega_0 t)$$
$$\sin(\omega_0 t + \omega_0\tau))$$

$$= E(AB)E(\cos(\omega_0 t)\cos(\omega_0 t + \omega_0\tau)) - E(A^2)E(\cos(\omega_0 t)\sin(\omega_0 t + \omega_0\tau))$$
$$+ E(B^2)E(\sin(\omega_0 t)\cos(\omega_0 t + \omega_0\tau)) - E(AB)E(\sin(\omega_0 t)\sin(\omega_0 t$$

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No.

$$\sigma_A^2 = E(A^2) - (E(A))^2 \to 0$$

$$\sigma_B^2 = E(B^2) - (E(B))^2 \to 0$$

But $\sigma_A^2 = \sigma_B^2 = \sigma^2 = E(A^2) = E(B^2)$

$$= -\sigma^2 E(\cos(\omega_0 t)\sin(\omega_0 t + \omega_0 \tau)) + \sigma^2 E(\sin(\omega_0 t)\cos(\omega_0 t + \omega_0 \tau))$$

$\sin(a-b) = \sin a \cos b - \cos a \sin b$

$$= \sigma^2 E(\sin(\omega_0 t - \omega_0 t - \omega_0 \tau))$$

$$= -\sigma^2 E(\sin(\omega_0 \tau))$$

$$= -\sigma^2 \sin(\omega_0 \tau)$$

$$= R_{xy}(\tau)$$

So, $X(t)$, $Y(t)$ are jointly wide-sense stationary. (#)

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No.

⊛ Covariance function :-

- The definition of covariance function can be extended to random process.

↝ The auto-covariance function is

$$C_{xx}(t, t+\tau) = E(x(t)\, x(t+\tau)) \overset{*}{=} E(x(t))\, E(x(t+\tau))$$
$$= R_{xx}(t, t+\tau) - E(x(t))\, E(x(t+\tau)).$$

↝ The cross covariance function is :-

$$C_{xy}(t, t+\tau) = E(x(t)\, y(t+\tau)) - E(x(t))\, E(y(t+\tau))$$
$$= \underbrace{R_{xy}(t, t+\tau) - E(x(t))\, E(y(t+\tau))}_{\text{Cross-correlation.}}$$

⊛ If $x(t)$, $y(t)$ are wide-sense stationary, then

$$C_{xx}(t, t+\tau) = R_{xx}(t, t+\tau) - \bar{x}^2 = R_{xx}(\tau) - \bar{x}^2$$
$$C_{xy}(t, t+\tau) = R_{xy}(t, t+\tau) - \bar{x}\,\bar{y}$$

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No.

* If $X(t), Y(t)$ are jointly wide sense stationary

$$C_{xx}(t, t+\tau) = R_{xx}(\tau) - \bar{X}^2$$
$$C_{yy}(t, t+\tau) = R_{yy}(\tau) - \bar{Y}^2$$
$$C_{xy}(t, t+\tau) = R_{xy}(\tau) - \bar{X}\bar{Y}$$

* For any random process, the variance can be calculated by

$$C_{xx}(t, t+\tau)\Big|_{\tau=0}$$

مهم

* Recall:-

$$Cov(X, Y) = E(XY) - E(X)E(Y)$$

→ Properties:-

→ $Cov(X, X) = E(XX) - (EX)(EX)$

$$= EX^2 - (EX)^2$$

$$= VarX = \sigma_X^2$$

Now:

$$C_{xx}(t, t+\tau) = E(x(t)\,x(t+\tau)) - E(x(t))\,E(x(t+\tau))$$

$$C_{xx}(t, t+\tau)\Big|_{\tau=0} = E(x(t)\,x(t)) - E(x(t))\,E(x(t))$$

$$= E(x^2(t)) - (E(x(t)))^2$$

$$= Var(x(t)).$$

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No.

(*) For any random process $x(t)$, $y(t)$,
if $x(t)$, $y(t)$ are indep,
then $C_{xy}(t, t+\tau) = 0$

(*) Discrete-time random process:-

It is defined as $x(t) = x(nT_s)$; $n = 0, \pm 1, \pm 2$

~> If $t_1 = n_1 T_s$, $t_2 = n_2 T_s$

$\rightarrow \tau = t_2 - t_1$,

$= n_2 T_s - n_1 T_s = (n_2 - n_1) T_s = \underline{K t_s}$

Here, the differences τ can only have discrete values KT_s.

So, for a DT-random process, we have mean
$= E(x(nT_s))$

auto-correlation $= R_{xx}(t, t+\tau)$

$= R_{xx}(nT_s, nT_s + KT_s)$

$= E(x(nT_s) X(nT_s + KT_s))$

Cross-correlation =

$R_{xy}(t, t+\tau) = R_{xy}(nT_s, nT_s + KT_s)$

$= E(x(nT_s)\, y(nT_s + KT_s))$

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No.

* auto-Covarian

* auto-Covariance $= C_{XX}(t, t+\tau)$
$= R_{XX}(nT_s, nT_s + kT_s) - E(x(nT_s))\, E(x(nT_s + kT$

- Cross-Corariance $= C_{xy}(t, t+\tau)$
$= R_{xy}(nT_s, nT_s + kT_s)$
$- E(x(nT_s))\, E(y(nT_s + kT_s))$

* For process that are wide-sense stationary

$E(x(nT_s)) = \bar{x}$, $E(y(nT_s)) = \bar{y}$

$R_{XX}(nT_s, nT_s + \underbrace{kT_s}_{\tau}) = R_{xx}(\underbrace{kT_s}_{\tau})$

$R_{yy}(nT_s, nT_s + kT_s) = R_{yy}(kT_s)$

$C_{xx}(nT_s, nT_s + kT_s) = R_{xx}(kT_s) - \bar{x}^2$

$C_{yy}(nT_s, nT_s + kT_s) = R_{yy}(kT_s) - \bar{y}^2$

$C_{xy}(nT_s, nT_s + kT_s) = R_{xy}(kT_s) - \bar{x}\bar{y}$

Joint

Scanned with CamScanner

Scanned with CamScanner